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Introduction of Amyloid-Related Imaging Abnormalities (ARIA) to the Brain

Third Year Individual Project – Final Report

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Contents

Contents 2

Declaration of originality 8

Intellectual property statement 9

Acknowledgements 10

1 Introduction 11

1.1 Background and motivation 11

1.2 Aims and objectives 12

1.3 Report Structure 13

2 Literature review 13

2.1 Brain Anatomy 13

2.2 ARIA 15

2.2.1 Introduction and Definition 15

2.2.2 Pathological Mechanisms of ARIA-E 16

2.2.3 Pathological Mechanisms of ARIA-H 17

2.2.4 Clinical Presentation, Detection and Management 18

2.3 Portable Gray Map Data and the Digital Human Phantom 20

2.3.1 Portable Gray Map Data 20

2.3.2 Digital Human Phantom 20

3 Methodology 21

3.1 Programming Language 21

3.2 Mathematical Models 22

3.2.1 ARIA-E 22

3.2.2 ARIA-H 23

3.3 White Matter and Cerebral Cortex Masks 23

3.3.2 White Matter Mask. 24

3.3.3 Cerebral Cortex Mask 24
3.4 Code Implementation 25
3.4.1 Load DHP Function 25
3.4.2 ARIA-E Generation Functions 26
3.4.3 ARIA-H Generation Functions 27
3.4.4 PGM File Extraction 28
4 Results and discussion 29
4.1 Introduction 29
4.2 Relevance 29
4.2.1 Effusion 29
4.2.2 Edema 32
4.2.3 Microhaemorrhages 35
4.2.4 Superficial Siderosis 38
4.3 Efficiency Analysis 40
4.3.1 Load DHP Function 40
4.3.2 Lesion Generation Functions 41
4.3.3 Feasibility 42
4.4 Effectiveness and Impact 43
4.5 Future Works 44
5 Conclusions 45
References 46
Appendices 50

Word count: 8240

List of Figures

Figure 1.1- Main Characteristics of ARIA, Risk Factors and Potential Predictors 12
Figure 2.1- poles and lobes of the cerebral hemisphere 14
Figure 2.2- neuroanatomy of the brain with respect to white matter, grey matter and cerebrospinal fluid 15
Figure 2.3- Formation and Deposition of Aβ plaques in blood vessels of the Brain 16
Figure 2.4- Differences in Manifestations of Effusion and Edema 16
Figure 2.5- Differences in Manifestations of Microhaemmorhages and Superficial Siderosis 17
Figure 2.6- MRI Scans of Effusion and Edema at Mild, Moderate, Severe Levels 19
Figure 2.7- MRI Scans of Microhaemmorhages at Mild, Moderate, Severe Levels and Superficial Siderosis at Mild and Moderate Levels 19
Figure 2.8- PGM Image Example 20
Figure 2.9- Healthy DHP Axial, Sagittal and Coronal Slices 21
Figure 3.1- MRI vs Generated Model of a Healthy Brain. 24
Figure 3.2- MRI vs Generated Model of a Brain. 25
Figure 4.1- MRI vs Generated Model of Mild Effusion. 30
Figure 4.2- MRI vs Generated Model of Moderate Effusion. 31
Figure 4.3- MRI vs Generated Model of Severe Effusion. 32
Figure 4.4- MRI vs Generated Model of Mild Edema. 33
Figure 4.5- MRI vs Generated Model of Moderate Edema. 34
Figure 4.6- MRI vs Generated Model of Severe Edema. 35
Figure 4.7- MRI vs Generated Model of Mild Microhaemorrhage. 36
Figure 4.8- MRI vs Generated Model of Moderate Microhaemorrhage. 37
Figure 4.9- MRI vs Generated Model of Severe Microhaemorrhage. 38
Figure 4.10- MRI vs Generated Model of Mild Siderosis. 39
Figure 4.11- MRI vs Generated Model of Moderate Superficial Siderosis. 40
Figure 4.12- Mean Execution Time of the Load DHP Function. 41
Figure 4.13- Mean Execution Time of the Lesion Generation Functions. 42
Figure 5.1- T2 FLAIR and T2*-GRE MRI images for Visualisation. 58

List of Tables

Table 1- Summary of ARIA-E and ARIA-H severity levels and features. 18

Table 2- PGM Image Structure.....20

List of Abbreviations

AD	Alzheimer's Disease
Aβ	Amyloid-beta
mAbs	Monoclonal Anti-bodies
ARIA	Amyloid-Related Imaging Abnormalities
ARIA-E	Amyloid-Related Imaging Abnormalities- Edema
ARIA-H	Amyloid-Related Imaging Abnormalities- Haemorrhages
MRI	Magnetic Resonance Imaging
AI	Artificial Intelligence
CAA	Cerebral Amyloid Angiopathy
BBB	Blood-Brain Barrier
PGM	Portable Gray Map
DHP	Digital Human Phantom
CSF	Cerebrospinal Fluid
EM	Electromagnetic
T1	T1-Weighted
T2	T2-Weighted
FLAIR	Fluid-Attenuated Inversion Recovery
GRE	Gradient-Recalled Echo
SWI	Susceptibility-Weighted Imaging
I/O	Input/Output
SSIM	Structural Similarity Index Measure
DWI	Diffusion-Weighted Imaging

Abstract

Alzheimer's disease is a major health issue and the leading cause of dementia, affecting millions of individuals worldwide. While emerging therapies such as Lecanemab can slow disease progression during the stages of subjective cognitive decline or early mild cognitive impairment, they can also lead to complications such as Amyloid-Related Imaging Abnormalities. Regular MRI monitoring is essential for detecting these abnormalities, but this process is both time consuming and costly, requiring expert analysis. Automating Amyloid-Related Imaging Abnormalities detection through Artificial Intelligence offers a promising solution to improve efficiency; however, the limited availability of MRI images featuring Amyloid-Related Imaging Abnormalities presents a significant challenge for algorithm training.

This project addresses this gap by examining existing MRI images of Amyloid-Related Imaging Abnormalities, extracting their defining features, and translating them into mathematical models. These models are then integrated into a Digital Human Phantom to generate synthetic MRI data that accurately represents Amyloid-Related Imaging Abnormalities manifestations. By developing anatomically realistic lesions, this research aims to enhance Artificial Intelligence training datasets, facilitating the advancement of automated Amyloid-Related Imaging Abnormalities detection and improving diagnostic capabilities in clinical settings.

Declaration of originality

I hereby confirm that this dissertation is my own original work unless referenced clearly to the contrary, and that no portion of the work referred to in the dissertation has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

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1 Introduction

1.1 Background and motivation

Alzheimer's disease (AD) is the most common neurodegenerative disorder in the world. It is characterised by progressive cognitive decline, memory loss and behavioural impairments, leading to loss of independence and death [1]. The pathophysiology of AD is primarily associated with the accumulation of amyloid-beta (Aβ) plaques and tau protein tangles, which disrupt neuronal function and lead to widespread neurodegeneration [2][3]. The burden of AD is increasing, with cases expected to triple by 2050 [4]. Despite no current cure for the disease, several treatments, meant to be administered during the early stages, are being developed to slow its progression.

Recent advancements in Alzheimer's therapy have focused on anti-amyloid monoclonal antibodies (mAbs), which target and clear amyloid-beta plaques. Among these, Lecanemab has demonstrated the most promise. In the Clarity AD clinical trial, Lecanemab showed a 25% reduction in cognitive decline compared to a placebo in patients with early AD, marking a significant breakthrough [5]. Taken intravenously every two weeks, Lecanemab works by binding to soluble and insoluble amyloid-beta aggregates, promoting their clearance by microglial cells [6]. Unlike earlier therapies, which faced setbacks due to lack of efficacy or severe side effects, Lecanemab has provided the first conclusive evidence of disease-modifying effects in early AD stages.

However, while these therapies offer new hope, they also introduce safety concerns in the form of Amyloid-Related Imaging Abnormalities (ARIA). ARIA is a complication associated with anti-amyloid therapies, commonly detected by Magnetic Resonance Imaging (MRI). It is broadly classified into:

- **ARIA-E (Edema/Effusion)** – Presents as focal or diffuse hyperintensities on FLAIR MRI due to vasogenic edema or sulcal effusion.
- **ARIA-H (Superficial Siderosis/Microhaemorrhages)** – Appears as hypointense signals on *T2-GRE or SWI MRI**, indicating microbleeds or superficial siderosis [7].

Figure 1.1 describes the main characteristics of ARIA, illustrating a comparison between the manifestations of ARIA-E and ARIA-H.

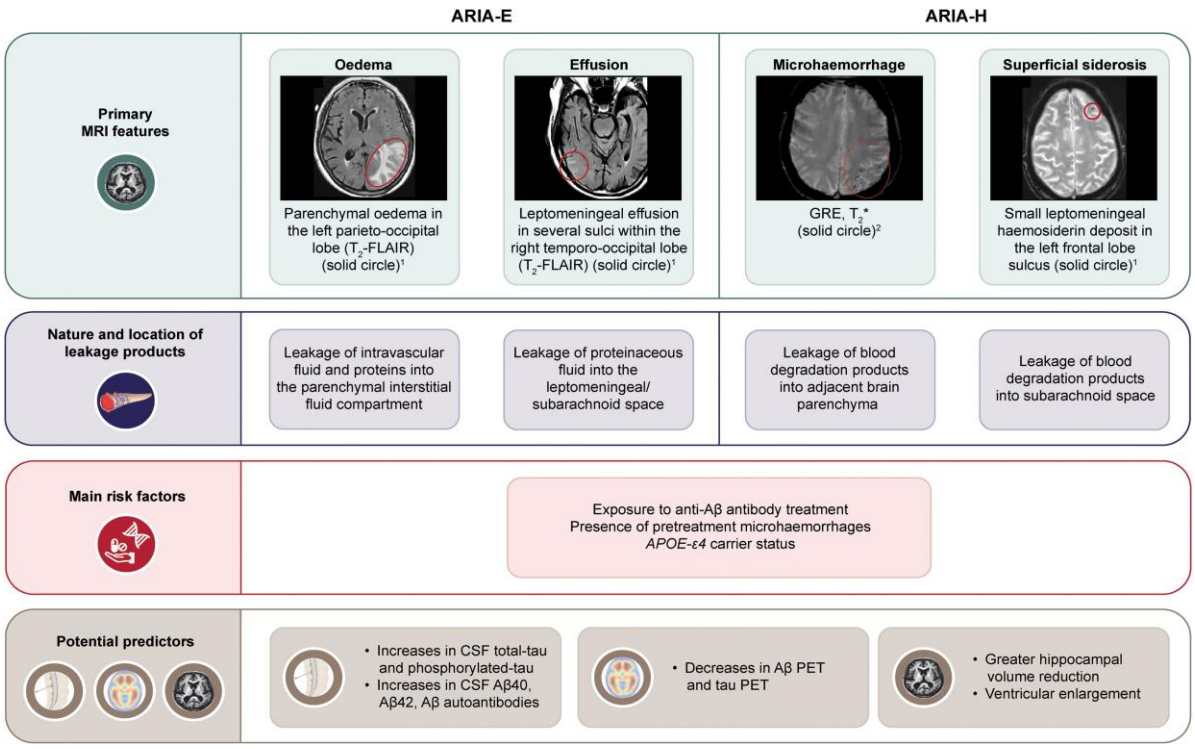


Figure 1.1- Main Characteristics of ARIA, Risk Factors and Potential Predictors [7].

These abnormalities are often asymptomatic, but in some cases manifest with headaches, confusion, seizures or stroke-like symptoms [8].

1.2 Aims and objectives

This project aims to develop computational models of Amyloid-Related Imaging Abnormalities, to generate synthetic MRI datasets for training artificial intelligence (AI) algorithms in automatic ARIA detection. These models will be mathematically defined and implemented within a Digital Human Phantom to ensure anatomical accuracy and improve the availability of training data for machine learning applications in neuroimaging. The specific objectives are as follows:

1) To analyse the clinical presentation and MRI characteristics of ARIA-E and ARIA-H, informing the design of realistic lesion models.
2) To examine the structure and segmentation of the Phantom and prepare it for the integration of simulated abnormalities.
3) To design and implement a method for embedding synthetic ARIA lesions into appropriate anatomical regions while maintaining structural fidelity.
4) To export the modified datasets in a format compatible with electromagnetic simulation and AI model development.

5) To assess the visual and structural quality of the simulated data and propose areas for future development.

1.3 Report Structure

- Chapter 2: Literature Review covering anatomy of the brain relevant to ARIA, ARIA introduction, pathological mechanisms and clinical presentation, Portable Gray Map Data and The Digital Human Phantom.
- Chapter 3: Methodology covering the choice of programming language for this project, mathematical models for ARIA-E and ARIA-H, implementation of white matter and cerebral cortex masks and the implementation of the code regarding loading the Human Phantom, generating the models and extracting the Portable Gray Map files.
- Chapter 4: Results and Discussion presenting the generated models and scientifically comparing them with MRI scans of real ARIA cases, evaluation of the efficiency of the program and its suitability to be used in generating mass synthetic models for algorithm training, effectiveness of the project and future works recommendations.
- Chapter 5: Conclusion wrapping up the report, summarising key findings, aligning them with the aims and objectives set at the start and emphasising future works.

2 Literature review

2.1 Brain Anatomy

The human brain is highly structured, with distinct regions organised to conduct specialised functions. At the broadest level, it consists of grey matter (neuron cell bodies) and white matter (myelinated nerve fibres) [9]. The grey matter forms the outer layer called the cerebral cortex, which is arranged in folds known as gyri and sulci [10]. The cerebral cortex is involved in high level brain functions such as sensory perception, cognition, and voluntary motor activities. The folding of the cortex increases its surface area for enhanced neural connectivity [11]. It is divided into four lobes: frontal, temporal, parietal and occipital lobes [12]; as illustrated in **Figure 2.1.**

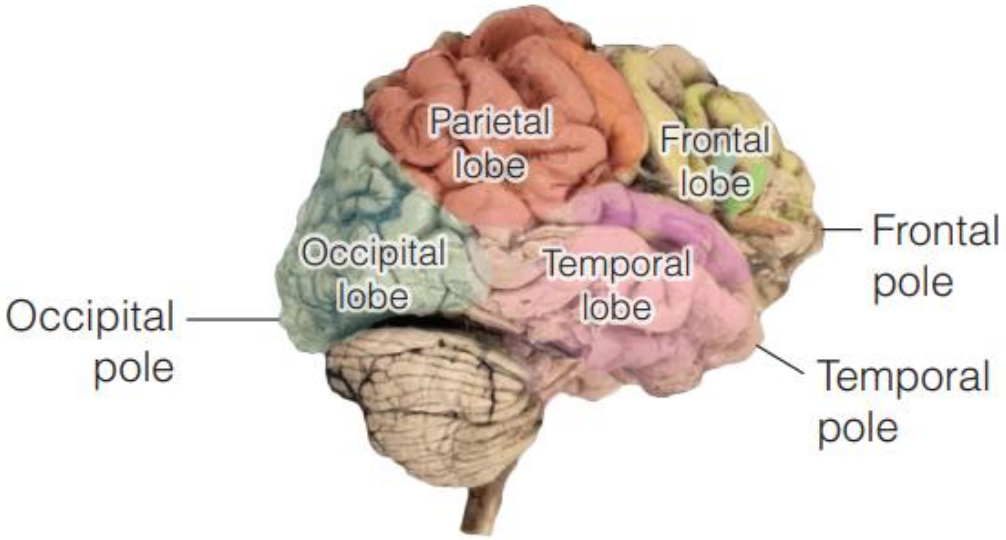


Figure 2.1- poles and lobes of the cerebral hemisphere [10].

The frontal lobe, positioned at the anterior portion of the cerebral cortex, is associated with decision-making, executive function, motor skills, and emotional regulation. Dysfunction or pathology in this area can result in cognitive deficits, behavioural changes, and motor impairment [13].

Located posterior to the frontal lobe, the parietal lobe integrates sensory information from various modalities, particularly spatial sense, navigation, and somatosensory processing. This area is crucial for attention and spatial awareness, often implicated in neurological diseases presenting with spatial disorientation [14].

Situated posteriorly within the cerebrum, the occipital lobe is primarily associated with visual processing. It houses the primary visual cortex, making it a key area in interpreting visual stimuli. Damage or abnormalities in this region often lead to visual disturbances or impaired visual interpretation [15].

The temporal lobe, located laterally, is significant for auditory processing, memory encoding, and language comprehension. It also plays an essential role in the formation and retrieval of memories and is commonly involved in conditions affecting memory such as Alzheimer's disease [16].

Beneath the cortical grey matter lies the cerebral white matter, consisting mainly of myelinated axons that facilitate communication between different brain regions and between the brain and the rest of the nervous system [17]. **Figure 2.2** visualises the placement of white matter with respect to the grey matter and cerebrospinal fluid. White matter is vital for efficient neural

communication and signal conduction and Pathologies involving cerebral white matter often disrupt cognitive and motor functions due to impaired information transfer [18].

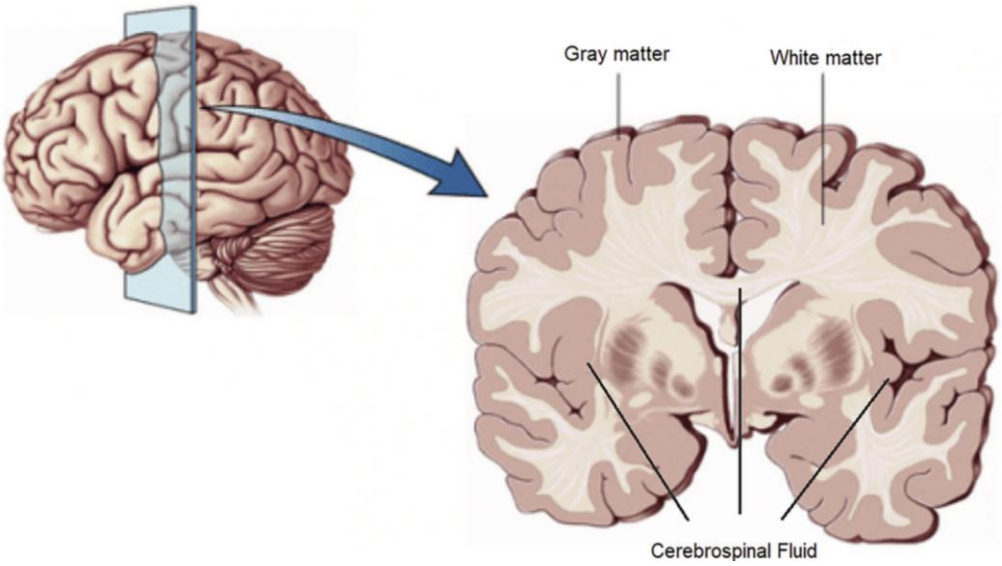


Figure 2.2- neuroanatomy of the brain with respect to white matter, grey matter and cerebrospinal fluid [17].

The cerebral vasculature, comprising arteries, veins, and capillaries, ensures the supply of oxygenated blood to the brain tissue and removal of metabolic waste. The integrity of this vascular network is critical. Any impairment, as seen in conditions such as cerebral amyloid angiopathy (CAA), significantly contributes to the development of ARIA. Amyloid deposits can lead to vascular fragility, leakage, and subsequent brain edema or haemorrhage, which are characteristic of ARIA [7].

2.2 ARIA

2.2.1 Introduction and Definition

Amyloid-Related Imaging Abnormalities describe characteristic neuroimaging changes observed in patients undergoing anti-amyloid therapies for Alzheimer's disease, particularly monoclonal antibodies targeting Aβ plaques [19]. ARIA comprises two primary subtypes: ARIA-E, representing vasogenic edema and sulcal effusion, and ARIA-H, which includes cerebral microhaemorrhages and superficial siderosis. ARIA closely correlates with CAA, a condition characterised by the deposition of Aβ in small to medium sized blood vessels of the brain as shown in **Figure 2.3**.

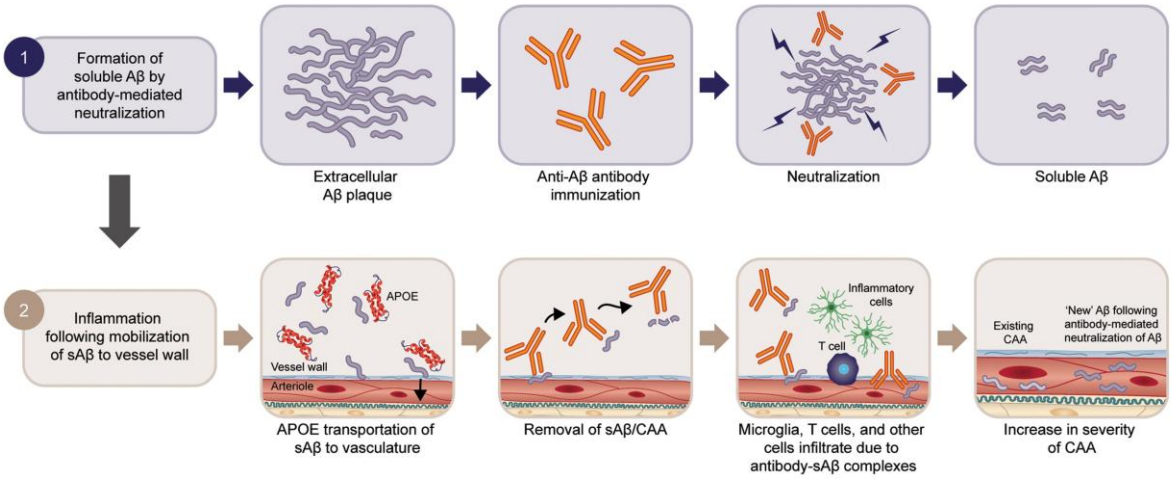


Figure 2.3- Formation and Deposition of Aβ plaques in blood vessels of the Brain (Adapted) [7].

This vascular amyloid deposition compromises vessel integrity, resulting in increased susceptibility to vascular leakage and haemorrhagic events [20]. First documented in early clinical trials, ARIA is increasingly recognised as a significant consideration in Alzheimer's treatment regimens [21].

2.2.2 Pathological Mechanisms of ARIA-E

ARIA-E is primarily characterised by two closely related manifestations: vasogenic edema and sulcal effusion. These phenomena stem from disruptions of the blood–brain barrier (BBB), which normally acts as a selective barrier preventing harmful substances in the bloodstream from entering the brain tissue [22]. **Figure 2.4** shows the differences in these manifestations.

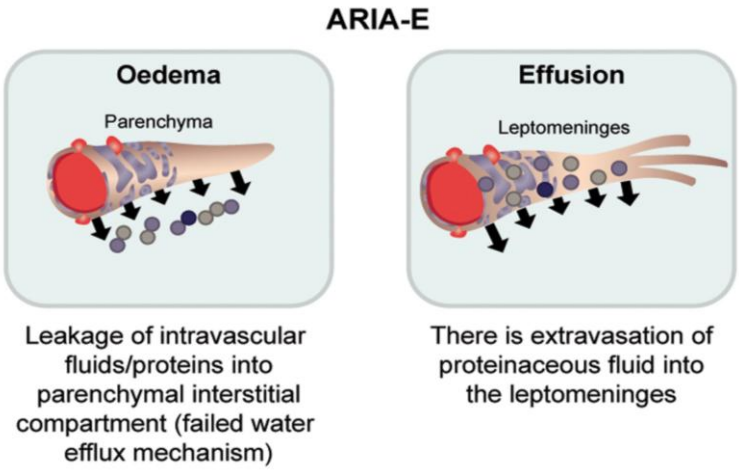


Figure 2.4- Differences in Manifestations of Effusion and Edema (Adapted) [7].

Sulcal effusion occurs with fluid accumulation specifically in the leptomeningeal spaces surrounding the brain. The same mechanism of increased vascular permeability allows protein-rich

fluid to escape blood vessels, but in this case, it accumulates within the cortical sulci and leptomeninges, creating fluid collections at the surface of the brain [23]. This fluid is termed proteinaceous due to its high protein content, distinguishing it from simple cerebrospinal fluid collections [24].

In contrast, edema involves the tight junctions of endothelial cells forming the BBB becoming compromised due to immune responses triggered by anti-amyloid therapies. These therapies cause inflammation and vessel wall weakening. The inflammatory response involves immune cells releasing inflammatory molecules and enzymes, such as matrix metalloproteinases, which further degrade the vessel wall structure [7]. Consequently, intravascular fluids and proteins leak from these weakened vessels directly into the extracellular spaces of the brain parenchyma, resulting in fluid accumulation between cells. This fluid accumulation is termed vasogenic edema and is characteristically reversible, as there is no cell death or permanent damage to the brain tissue [25].

2.2.3 Pathological Mechanisms of ARIA-H

ARIA-H comprises of two key pathological presentations: microhaemorrhages and superficial siderosis, with different manifestation processes illustrated by **Figure 2.5**. Both phenomena result primarily from vascular injury due to CAA.

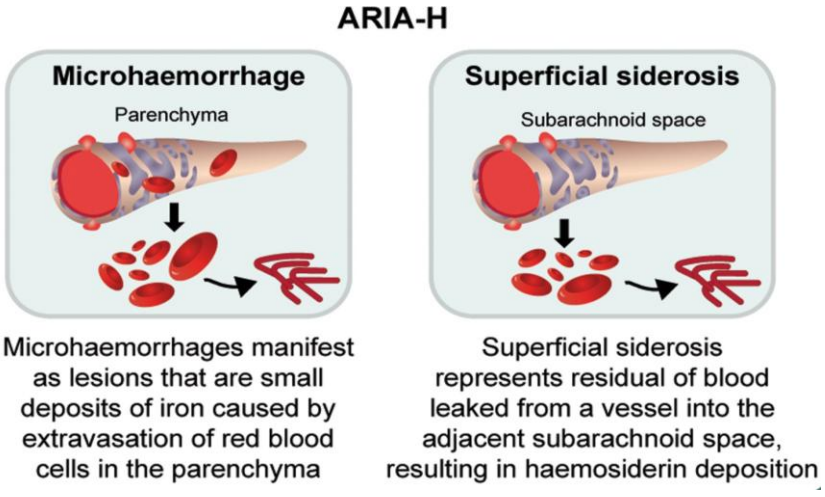


Figure 2.5- Differences in Manifestations of Microhaemmorhages and Superficial Siderosis (Adapted) [7].

Microhaemorrhages occur when amyloid-laden vessels become structurally weakened, losing integrity and elasticity. This fragility leads to small-scale ruptures, causing red blood cells to extravasate into the brain tissue [26]. These microbleeds leave behind deposits of iron-rich compounds, which appear on MRI scans as small, hypointense (dark) lesions, predominantly detected using susceptibility sensitive imaging techniques such as T2*-GRE [27]. The iron deposits

remain visible even after the bleeding has resolved, making microhaemorrhages persistent imaging markers for vascular injury [28].

In contrast, superficial siderosis involves bleeding from amyloid-affected vessels directly into the subarachnoid space, rather than into the parenchyma. Haemoglobin from the leaked blood is metabolised into haemosiderin, a byproduct of red blood cell breakdown, which deposits on the superficial surfaces of the brain [29]. These haemosiderin deposits similarly manifest as linear hypointense areas along the cortical surface and sulci on MRI scans, marking sites of prior bleeding events [25]. Superficial siderosis indicates chronic or recurrent bleeding episodes, typically reflecting more advanced amyloid vascular pathology [30]. Both microhaemorrhages and superficial siderosis reflect permanent, structural vascular damage that differs fundamentally from the reversible inflammatory leakage observed in ARIA-E. ARIA-H lesions serve as critical indicators of vascular integrity and potential predictors of further haemorrhagic risk, particularly in patients undergoing prolonged anti-amyloid therapy [19].

2.2.4 Clinical Presentation, Detection and Management

ARIA often presents asymptomatically and is typically identified through routine MRI surveillance during therapeutic interventions. Symptomatic cases may present with mild-to-moderate neurological symptoms such as headaches, confusion, or dizziness. Severe ARIA manifestations can include seizures or raised intracranial pressure [31]. ARIA severity is classified based on lesion size and distribution, ranging from mild, moderate and severe classifications. Table 1 summarises the severity levels of ARIA and their corresponding features:

Table 1- Summary of ARIA-E and ARIA-H severity levels and features.

Severity Level	ARIA-E (Effusion and Edema)	ARIA-H (Microhaemorrhages and Superficial Siderosis)
Mild	Single lesion, localised edema or effusion (<5 cm)	Single or few microhaemorrhages (<5) or focal siderosis
Moderate	Multiple lesions or edema/effusion (5–10 cm)	Multiple microhaemorrhages (5–10) or multiple regions of siderosis
Severe	Extensive (>10 cm) edema or effusion	Extensive microhaemorrhages (>10) or widespread siderosis

Routine MRI is essential for ARIA detection during anti-amyloid therapies. Standard imaging protocols include baseline MRI scans, followed by regular imaging at intervals of 3, 6, 9, and 12

months. Detection relies on careful analysis by experienced radiologists, although issues such as costs and intensive labour are challenges [32][33]. **Figures 2.6** and **2.7** provide a visualisation of the MRI images depicting ARIA-E and ARIA-H cases at varying severity levels.

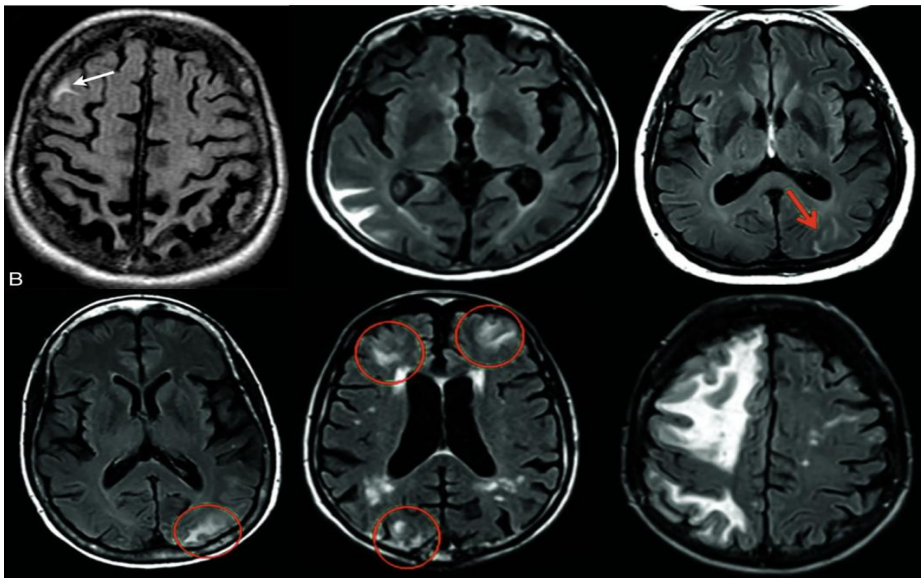


Figure 2.6- MRI Scans of Effusion (top) and Edema (bottom) at Mild, Moderate, Severe Levels (left to right) [25][19].

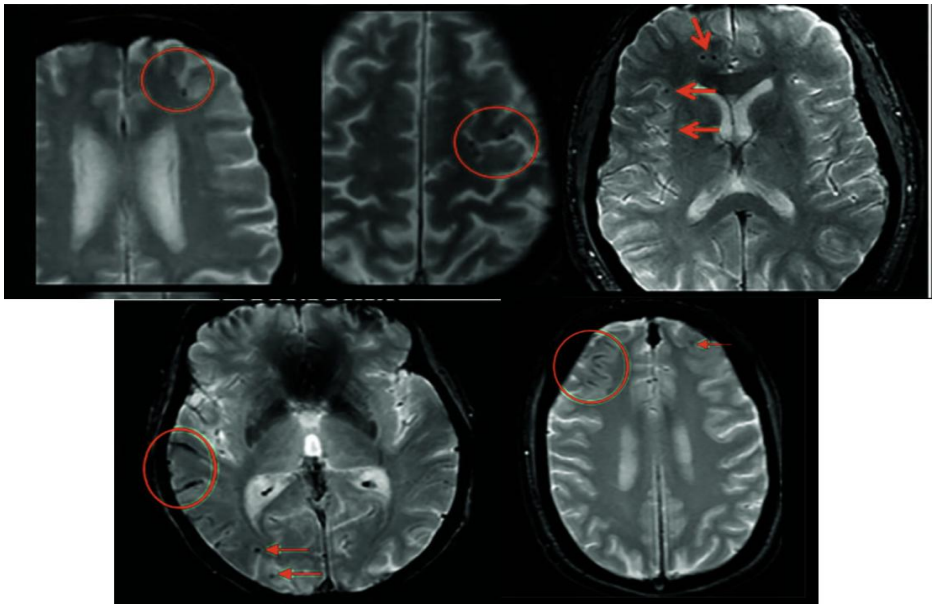


Figure 2.7- MRI Scans of Microhaemmorhages (top) at Mild, Moderate, Severe Levels (left to right) and Superficial Siderosis (bottom) at Mild and Moderate Levels (left to right) [19].

Management strategies depend upon severity and clinical presentation. Mild, asymptomatic ARIA typically requires monitoring without active intervention. Moderate to severe cases may necessitate temporary or permanent cessation of anti-amyloid therapy or corticosteroid

administration to reduce edema [34]. No specific treatment exists for ARIA-H, and therapeutic decisions balance ARIA associated risks against potential therapeutic benefits.

2.3 Portable Gray Map Data and the Digital Human Phantom

2.3.1 Portable Gray Map Data

Portable Gray Map (PGM) is a simple image file format commonly used for representing grayscale images. Due to its simplicity, it is widely utilised in research involving medical image processing and computational modelling [35]. A PGM file is structured in ASCII or binary encoding, with its structure shown in Table 2 accompanied by an example of a PGM image in **Figure 2.8**.

Table 2- PGM Image Structure.

Line Number	Description
1	Magic Number (file format)
2	Optional comment, begins with '#'
3	Width and Height Dimension of Pixels
4	Maximum Pixel Value
5	Pixel Intensity Values

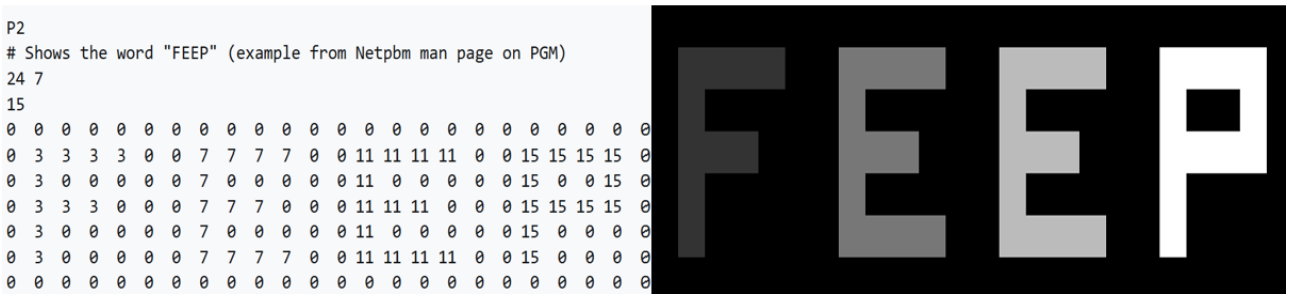


Figure 2.8- PGM Image Example [35].

2.3.2 Digital Human Phantom

A Digital Human Phantom (DHP) is a detailed numerical model of human anatomy, which accurately represents biological tissues and structures numerically. The DHP used in this project was provided by RIKEN (Saitama, Japan) under a no-disclosure agreement between RIKEN and the University of Manchester. The usage was approved by RIKEN ethical committee. The DHP enables realistic simulations of physical phenomena, including electromagnetic (EM) wave propagation,

which are critical in diagnostic imaging and treatment research [36]. The DHP data can be segmented into three anatomical planes: axial, sagittal, and coronal; as illustrated by **Figure 2.9**.



Figure 2.9- Healthy DHP Axial, Sagittal and Coronal Slices (left to right).

Each plane provides unique cross-sectional views that facilitate targeted analysis and modelling. In this project, the axial plane is exclusively used due to its common application in brain imaging, notably MRI scans for ARIA detection.

3 Methodology

3.1 Programming Language

The selection of an appropriate programming language was pivotal for the successful implementation of the project, as it influenced the development, efficiency, and assessment of the lesion models. MATLAB was considered along other languages due to several significant advantages. Firstly, MATLAB offers extensive ease of use and flexibility in customising code, particularly beneficial when creating and refining multiple lesion models across different anatomical regions of the brain. Also, its powerful built-in functions and intuitive scripting allow rapid prototyping and straightforward assessment of the generated models.

However, MATLAB presented notable drawbacks in the context of image handling and efficiency. A major issue is MATLAB's inherent autoscaling feature when generating PGM files. Autoscaling unintentionally converts the integer pixel intensities to floating point numbers, requiring additional processing steps to revert the data to the correct integer format. This introduces complications to the workflow for generating valid PGM files suitable for future EM wave propagation simulations. Additionally, MATLAB exhibits significant computational overhead, which

stems from its internal memory management and higher-level abstraction, resulting in slower execution compared to languages closer to machine level.

Alternatives to MATLAB such as Fortran, C, and Python were considered as potential solutions to these problems. Fortran and C, renowned for their computational efficiency and lower memory overhead, directly address MATLAB's efficiency shortcomings by offering rapid execution and minimal system overhead, essential for large-scale simulations. Python, while higher-level than C or Fortran, also presents distinct advantages with powerful numerical libraries such as 'NumPy' and 'SciPy' that facilitate efficient data handling without unintended autoscaling. Python further benefits from widespread community support, enabling easier troubleshooting and code optimisation.

Despite these advantages, MATLAB was ultimately selected primarily due to familiarity and extensive experience with the language. This choice allowed more efficient project development within the given timeframe, enabling rapid iterative design and evaluation of lesion models. Nevertheless, it is acknowledged that future works should consider adopting more efficient languages like Python or C to mitigate the highlighted limitations and enhance performance and applicability in research scenarios.

3.2 Mathematical Models

3.2.1 ARIA-E

The Gaussian distribution was selected to model ARIA-E lesions. The Gaussian function effectively captures the gradual intensity changes and diffuse boundaries characteristic of fluid accumulation observed clinically in ARIA-E. This mathematical approach ensures biologically accurate representation and enables precise control over lesion morphology. The Gaussian distribution employed for ARIA-E is mathematically defined as:

$$\text{ARIA-E}\,(x, y, z) = I_{peak} \cdot e^{-\left(\frac{(x-x_c)^2}{2\sigma_x{}^2} + \frac{(y-y_c)^2}{2\sigma_y{}^2} + \frac{(z-z_c)^2}{2\sigma_z{}^2}\right)} \quad (1)$$

Where I_{peak} is the peak intensity of the lesion, (x_c, y_c, z_c) are the central coordinates of the lesion with respect to the DHP and (σ_x, σ_y, σ_z) are the parameters controlling the spread of the lesion along each respective axis.

The distribution Realistically reproduces the smooth and diffuse appearance of edema and effusion and allows precise adjustment of lesion parameters for accurate anatomical positioning. This facilitates reproducible and consistent lesion creation across varying severity levels.

3.2.2 ARIA-H

ARIA-H is characterised by discrete and sharply defined hypointense lesions. To accurately represent these lesions, a Gaussian distribution with a binary threshold of '0.6' was adopted. This model was preferred over simple spherical or scattered pixel-based approaches as it effectively captures the irregular shapes and sharp boundaries typical of clinical haemorrhagic lesions.

The Gaussian-binary model for ARIA-H is mathematically expressed as:

$$\text{ARIA-H}\,(x, y, z) = I_{\text{lesion}} \cdot e^{-\left(\frac{(x-x_c)^2}{2\sigma_x^2} + \frac{(y-y_c)^2}{2\sigma_y^2} + \frac{(z-z_c)^2}{2\sigma_z^2}\right)} \geq 0.6 \qquad (2)$$

Where I_{lesion} is the intensity of the lesion, (x_c, y_c, z_c) are the central coordinates of the lesion with respect to the DHP and (σ_x, σ_y, σ_z) are the parameters controlling the spread of the lesion along each respective axis.

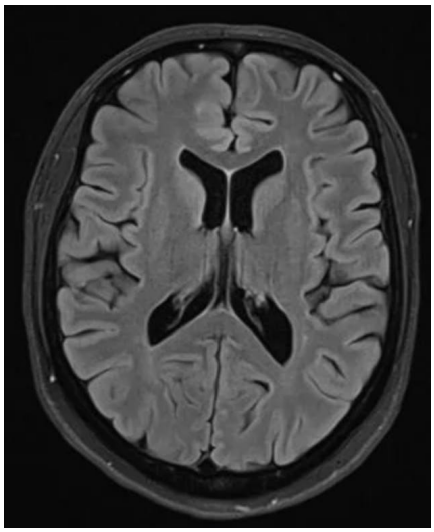
The binary threshold ensures the hypointense lesions are represented with a discrete intensity in the model, Ensuring precise anatomical localisation and clearly defined lesion boundaries. This approach also maintains the irregular shape of haemorrhagic lesions, although further improvements, as discussed in Chapter 4, could be made.

3.3 White Matter and Cerebral Cortex Masks

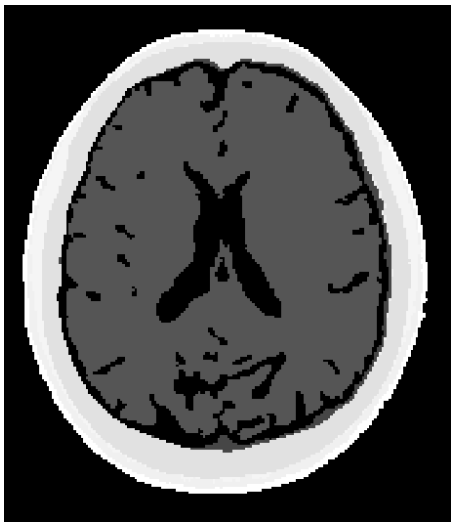
To ensure anatomical fidelity in the generation of synthetic ARIA models, this project employed binary tissue masks to define and constrain lesion placement within the DHP. Given that different ARIA subtypes affect specific brain regions it was necessary to segment the DHP into biologically meaningful domains. By isolating white matter and cortical grey matter through voxel intensity values, it became possible to simulate lesions in accordance with their clinical radiological presentations. These anatomical masks were therefore integral to preserving the structural integrity of the brain model and ensuring that synthetic abnormalities reflected the spatial patterns observed in MRI scans.

3.3.2 White Matter Mask.

The white matter mask was constructed by isolating the white matter voxels from the DHP based on predefined pixel intensity values provided by RIKEN. In the version of the DHP used in this project, white matter regions were encoded with a voxel intensity of '2'. This allowed for straightforward segmentation using logical operations, producing a binary mask that could be applied across all slices in the three-dimensional brain volume. **Figure 3.1** compares a T2 FlAIR MRI scan of a healthy brain **(a)** to an axial slice of the DHP showing the resulting white matter mask **(b)**. The generated model preserves the spatial location and topology of the white matter while excluding adjacent structures such as the cerebral cortex, ventricles, and sulci.



(a) T2 FLAIR MRI of a Healthy Brain [37].



(b) Healthy Brain DHP Slice.

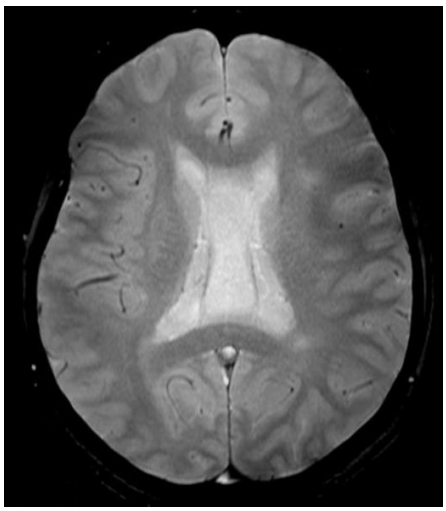
Figure 3.1- MRI vs Generated Model of a Healthy Brain.

The implementation began by scanning the DHP volume for all voxels equal to the white matter intensity value. This produced a binary mask where voxels belonging to white matter were marked as true, and all other regions—such as grey matter, CSF, and skull—were excluded. This binary structure was then given a pixel value of '17' and used as a spatial filter during lesion generation to ensure that abnormalities were placed only within designated white matter regions.

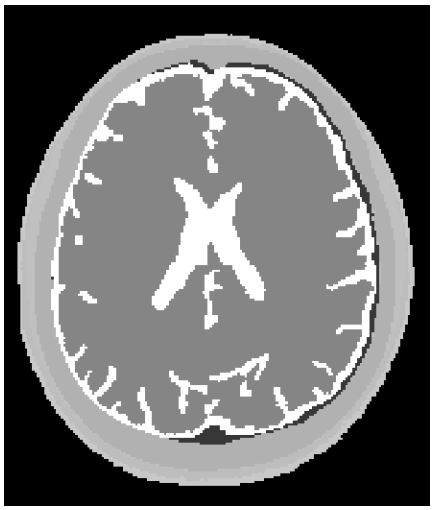
3.3.3 Cerebral Cortex Mask

The cerebral cortex mask was created by identifying voxels within the DHP corresponding to the cortical mantle, defined by a unique voxel intensity value of '1'. These regions typically represent cortical grey matter and the surrounding sulcal spaces in the DHP dataset. Using the same logical indexing technique as the white matter mask, these voxel locations were extracted into a binary

mask that delineated the full extent of the cortex across the three-dimensional volume and assigned a pixel intensity of '60'. **Figure 3.1** compares a T2*-GRE MRI scan of the brain **(a)**, to an axial slice of the DHP showing the resulting cerebral cortex mask **(b)**.



(a) T2*-GRE MRI of the Brain [38].



(b) Brain DHP Slice.

Figure 3.2- MRI vs Generated Model of a Brain.

3.4 Code Implementation

3.4.1 Load DHP Function

The DHP data used in this project was supplied in the form of sequential PGM image slices, each representing an axial cross-section of the brain. To reconstruct the DHP as a volumetric dataset suitable for lesion integration, a script was implemented to systematically load and compile these slices into a three-dimensional matrix.

The process began by identifying the directory containing the PGM files and generating a list of all available images. The dimensions of the DHP slices were inferred from the header information of the first file in the sequence, under the assumption that all slices shared a consistent resolution. This information was used to pre-allocate a three-dimensional array, 'DHP_3D', with dimensions corresponding to image height, width, and number of slices. Pre-allocation was necessary to ensure computational efficiency and avoid the memory reallocation overhead associated with dynamic array expansion.

Each PGM file was then loaded individually using low-level file operations. Header lines specifying the file format, comments, and metadata were skipped, and the image pixel values were read as numerical data. Since PGM files store pixel values column-wise, the data array was transposed

prior to insertion into the final 3D matrix to maintain anatomical orientation consistency with standard medical imaging conventions.

This method produced a volumetric DHP array in which each voxel retained its original pixel intensity value, representing a specific brain tissue class. By reconstructing the DHP in this way, the code provided a spatially accurate, layered representation of human brain anatomy, forming the foundational structure for all subsequent lesion modelling and image export.

3.4.2 ARIA-E Generation Functions

ARIA-E lesions were inserted into the DHP using a structured masking approach to ensure biological accuracy. Lesions were generated using the parameterised 3D Gaussian models, with severity and spatial extent modulated by the peak intensity and standard deviations along each axis. Both effusion and edema lesions were modelled in this manner but differed in their anatomical targets and insertion order.

Effusion was simulated by applying the lesion volume directly to the original DHP. Lesions were intentionally limited to the cortical areas by design. After lesion application, the white matter mask was reapplied as an overlay, this post-processing step ensured that the synthetic effusion did not overwrite any white matter structure, maintaining the spatial accuracy and biological realism of the model using the indexing of equation 3:

$$D(x,y,z) = \begin{cases} L_{effusion}(x,y,z), & if\ DHP_{original}(x,y,z) = 1 \\ 17, & if\ DHP_{original}(x,y,z) = 2 \\ DHP_{original}(x,y,z), & otherwise \end{cases} \quad (3)$$

Where D(x,y,z) is the DHP with the inserted lesions and white matter mask, $L_{effusion}(x,y,z)$ represents the effusion lesions applied to the cerebral cortex areas, $DHP_{original}(x,y,z)$ is the original DHP.

Edema lesions were also generated using Gaussian models but were constrained exclusively to the white matter. In contrast to effusion, the white matter mask was applied first, and lesion volumes were added on top of it, replacing only white matter voxels.

To represent irregular swelling patterns typical of vasogenic edema, multiple lesion volumes were generated, each with varied spatial parameters and positions. These were summed using equation 4 to form the final lesion map:

$$L_{edema_total}(x,y,z) = \sum_{i=1}^{n} L_i(x,y,z) \quad (4)$$

where L_{edema_total} Represents the final edema lesion after summing the lesion volumes.

The cumulative lesion volume was inserted only when the white matter mask was active, using the logical indexing in equation 5:

$$D(x,y,z) = \begin{cases} W(x,y,z) + L_{edema_total}(x,y,z), & if\ W(x,y,z) = 1 \\ DHP_{original}(x,y,z), & otherwise \end{cases} \quad (5)$$

Where W(x,y,z) represents the white matter mask, and all other variables keep their previously stated definitions.

The resulting DHP volume preserved all original tissue boundaries, with effusion lesions constrained to cortical structures and edema lesions strictly constrained to the white matter by reapplying the appropriate anatomical mask after each insertion. Final slice inspection using 'squeeze()' and 'imshow()' confirmed that lesion placement was accurate across selected axial planes.

3.4.3 ARIA-H Generation Functions

ARIA-H lesions representing microhaemorrhages and superficial siderosis were modelled using a discrete voxel-based approach. Lesions were inserted into the DHP by first restoring anatomical masks to preserve tissue integrity, followed by direct voxel overwrites at anatomically appropriate locations. Both lesion types were visually represented by a hypointense intensity value of '1', while the surrounding cerebral cortex mask was consistently assigned a value of '60' and the white matter mask was kept at a value of '17'.

Microhaemorrhages were represented as individual single-pixel lesions without volumetric spread, reflecting their punctate appearance in susceptibility-weighted imaging (SWI). Lesions were manually positioned within the white matter using predefined coordinates and scaled in number depending on severity.

To restrict microhaemorrhages to white matter, the white matter mask was first applied, and the lesion coordinates were then inserted as follows:

$$D(x,y,z) = \begin{cases} W(x,y,z) + L_{micro}(x,y,z), & if\ W(x,y,z) = 1 \\ 60, & if\ DHP_{original}(x,y,z) = 1 \\ DHP_{original}(x,y,z), & otherwise \end{cases} \quad (6)$$

Where L_{micro} represents the microhaemorrhage lesions and all other variables keep their previously stated definitions.

This ensured that microhaemorrhages were placed exclusively within white matter, and that surrounding regions were restored to their standard visual intensity for structural contrast.

Superficial siderosis lesions were modelled using broader thresholded Gaussian masks to simulate haemosiderin accumulation along the cortical surface. Each lesion was generated volumetrically, then binarized using a fixed threshold and applied exclusively to the cerebral cortex.

Before lesion insertion, the cortex mask was reapplied across the DHP at a fixed intensity of '60'. The binarized lesion volume was then used to overwrite cortex voxels with the same hypointense intensity value of '1':

$$D(x,y,z) \begin{cases} C(x,y,z) + L_{siderosis}(x,y,z), & if\ C(x,y,z) = 1 \\ W(x,y,z) + L_{micro}(x,y,z), & if\ W(x,y,z) = 1 \\ DHP_{original}(x,y,z), & otherwise \end{cases} \quad (7)$$

Where C(x,y,z) represents the cerebral cortex mask, $L_{siderosis}$ represents the superficial siderosis lesions and all other variables keep their previously stated definitions.

Superficial siderosis cases also included co-occurring microhaemorrhages. Therefore, following cortical insertion, the white matter mask was restored, and discrete microhaemorrhage voxels were applied using the same method as previously described. This two-phase process ensured proper spatial separation and biologically realistic lesion placement.

Final models were also validated visually by extracting axial slices through 'squeeze()' and 'imshow()' to confirm that both lesion types were confined to their respective tissue domains.

3.4.4 PGM File Extraction

A custom function was implemented to export the modified DHP slices as '.pgm' files, preserving their compatibility with external simulation frameworks. The function manually writes the ASCII-based PGM header, including the matrix dimensions and detected maximum pixel value, followed

by the pixel data row by row. Prior to export, the image matrix was explicitly cast to an unsigned 8-bit integer format to maintain formatting consistency and prevent unintended rescaling. This manual conversion bypassed MATLAB's default image-handling behaviour, which automatically normalises data and could distort the predefined intensity values assigned to anatomical structures and lesions. By preserving the original integer values, each voxel retained its intended diagnostic role and electromagnetic relevance within the simulation framework.

4 Results and discussion

4.1 Introduction

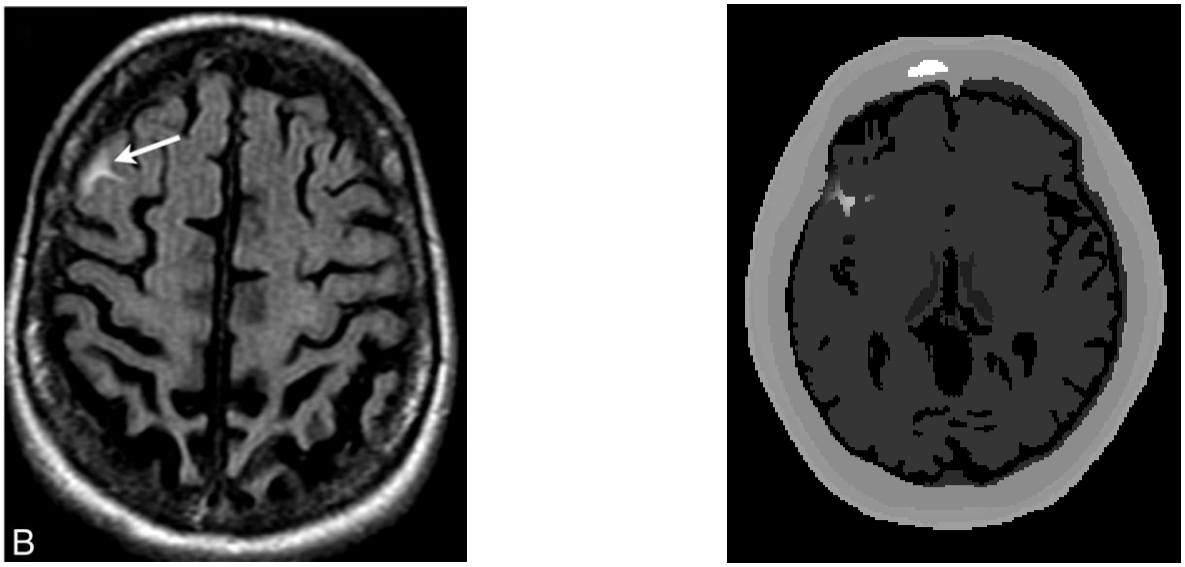
The algorithms introduced in chapter 3 were successful in generating lesions of different manifestations of ARIA-E and ARIA-H, however a thorough analysis must be completed to assess the accuracy of the lesions and their feasibility to be used as datasets in AI training. This chapter presents the results obtained from the algorithms and focuses on four pillars that assess them: Relevance, Efficiency, Effectiveness and Future Works.

4.2 Relevance

4.2.1 Effusion

1) Mild

Figure 4.1 (a) shows a FLAIR MRI scan that reveals a hyperintense signal in the sulci of the right frontal region, indicated by the arrow. This signal abnormality is characteristic of ARIA-E, suggesting the presence of fluid accumulation in the subarachnoid space. This case is classified as mild due to the localised nature of the sulcal hyperintensity, which is confined to a small region in the right frontal lobe without affecting multiple brain areas. Additionally, there is no evidence of mass effect, cortical compression, or significant white matter involvement, distinguishing it from more severe cases that exhibit widespread edema and increased intracranial pressure.



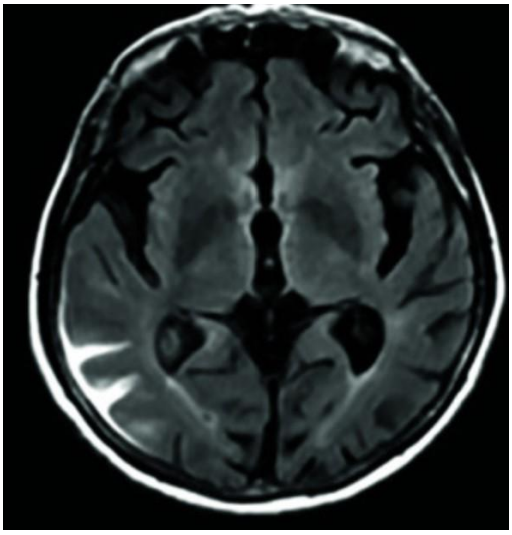
(a) MRI Scan presenting Mild Effusion [25]. **(b)** Model of Mild Effusion.

Figure 4.1- MRI vs Generated Model of Mild Effusion.

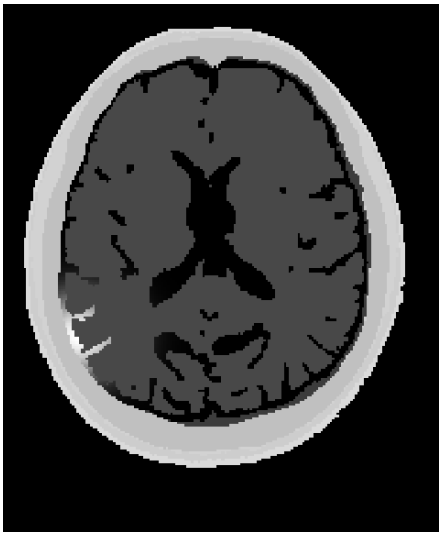
The generated ARIA-E model, shown in **Figure 4.1 (b)**, accurately reflects clinical cases by restricting hyperintensity to the sulcal regions of the cerebral cortex, aligning with the typical presentation of effusion in FLAIR MRI scans. The Gaussian distribution ensures a natural diffusion pattern, mimicking the fluid spread observed in real cases while maintaining smooth intensity transitions rather than abrupt changes. Additionally, by limiting the lesion to the cerebral cortex and avoiding white matter involvement, the model preserves biological accuracy, as ARIA-E primarily affects the subarachnoid space without infiltrating deeper brain structures.

2) Moderate

Figure 4.2 (a) demonstrates a moderate case of effusion, characterised by sulcal hyperintensity affecting the right posterior temporal and parietal lobes. The lesion, measuring 5–10 cm, exhibits a more extensive spread compared to mild cases, reflecting a greater degree of fluid accumulation within the subarachnoid space. The hyperintensity follows the anatomical contours of the sulci, consistent with ARIA-E pathology, while the surrounding brain structures remain intact, indicating the absence of significant mass effect or cortical compression. This pattern aligns with moderate ARIA-E presentations, which may remain asymptomatic or manifest with mild symptoms such as headaches or dizziness.



(a) MRI Scan Presenting Moderate Effusion [19].



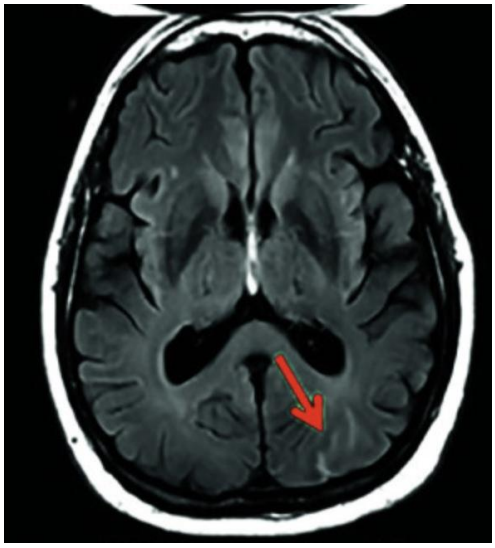
(b) Model of Moderate Effusion.

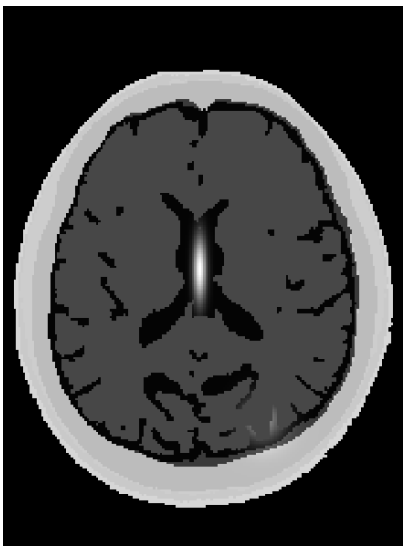
Figure 4.2- MRI vs Generated Model of Moderate Effusion.

The generated lesion model, illustrated by **Figure 4.2 (b)**, accurately captures the location and overall shape of the effusion, aligning with real cases by targeting the right posterior temporal and parietal lobes. The spread follows the sulcal contours, although the thickness of the effusion is slightly reduced due to variations in sulcal spacing within the Digital Human Phantom compared to the real MRI case. The Gaussian intensity gradient ensures a smooth diffusion effect, avoiding abrupt transitions and maintaining a realistic fluid accumulation pattern within the subarachnoid space, it is important to note that the generated lesion also diffuses along the edge of the brain without overwriting the skull and white matter.

3) Severe

The MRI image of severe effusion in **Figure 4.3 (a)** is characterised by extensive sulcal hyperintensity affecting the parieto-occipital region, with fluid accumulation spanning multiple sulci. The signal intensity indicates a substantial effusion volume, and the lesion's thickness surpasses mild and moderate cases. The diffuse spread across several cortical folds rather than a localised area highlights the severity of the condition, as larger effusions pose a greater risk of sulcal widening and mass effect, which can contribute to neurological symptoms. This case may necessitate clinical intervention due to the increased likelihood of patient symptoms and complications. **Figure 4.3 (b)**, showing the generated lesion, simulates the extensive sulcal effusion observed in clinical MRI scans.





(a) MRI Scan Presenting Severe Effusion [19]. **(b)** Model of Severe Effusion.

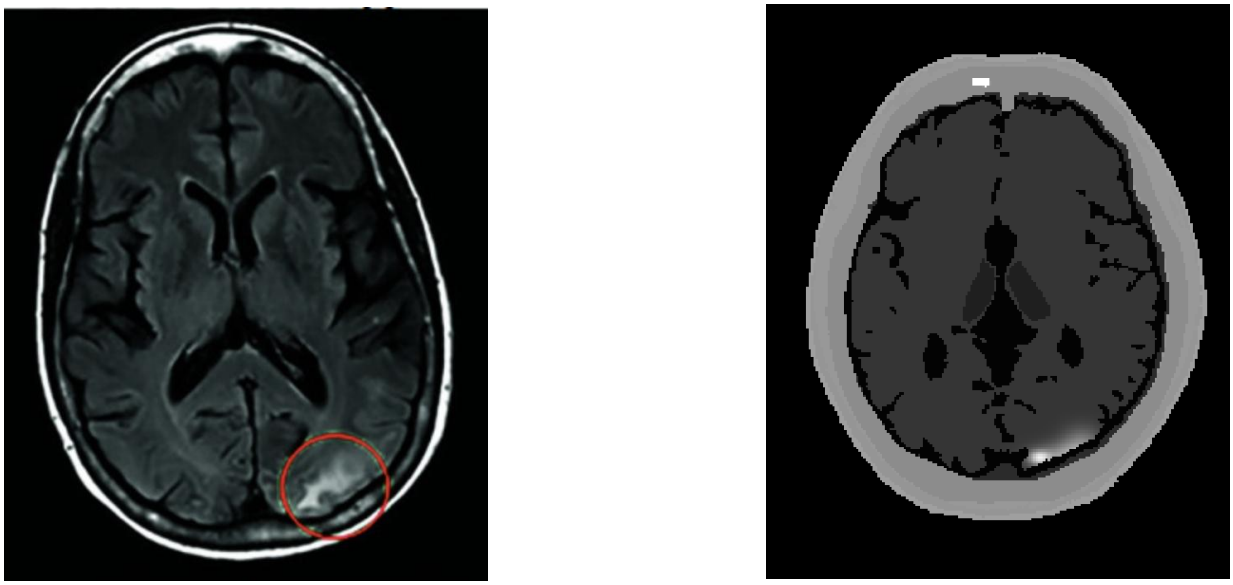
Figure 4.3- MRI vs Generated Model of Severe Effusion.

The model accurately replicates the substantial fluid accumulation within the subarachnoid space. The lesion's distribution across multiple sulci in the parieto-occipital region aligns with typical severe effusion presentations. However, the differences between the brain structures in the MRI scan and DHP result in variations in sulcal spacing and morphology, leading to minor deviations in lesion thickness and precise localisation. To enhance the model's accuracy, further refinements could involve incorporating more detailed anatomical variations and adjusting the intensity gradients to better mimic the observations made in clinical cases.

4.2.2 Edema

1) Mild

The MRI scan for this case, illustrated by **Figure 4.4 (a)**, reveals a hyperintense signal in the left parieto-occipital subcortical white matter, measuring less than 5 cm in the transverse plane. The lesion exhibits mild local mass effect, indicated by fluid accumulation without significant structural displacement. The restricted spread and absence of extensive cortical involvement are consistent with mild edema. No haemorrhagic transformation or secondary abnormalities are observed on this sequence. The generated mild ARIA-E lesion model effectively reproduces the T2-FLAIR hyperintensity observed in clinical cases, as shown in **Figure 4.4 (b)**, localised to the left parieto-occipital subcortical white matter.



(a) MRI Scan Presenting Mild Edema [19]. **(b)** Model of Mild Edema.

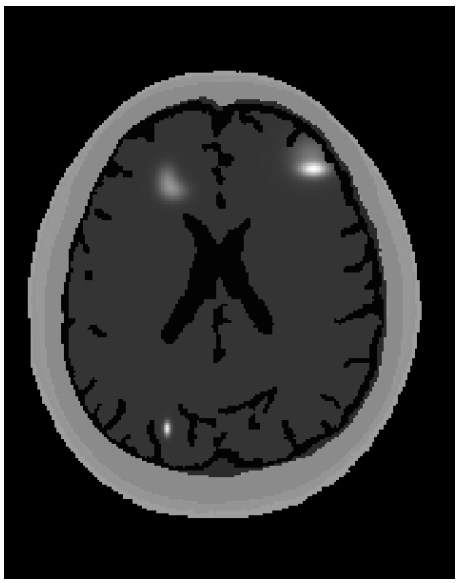
Figure 4.4- MRI vs Generated Model of Mild Edema.

The lesion was modelled using multiple Gaussian distributions at varying intensities, ensuring a realistic diffusion pattern that mimics the gradual fluid spread characteristic of vasogenic edema. The placement is restricted to the white matter, preventing overwriting of the cerebral cortex or CSF regions, thereby maintaining biological accuracy. The lesion extends along the white matter border with a decreasing intensity gradient, simulating the progressive dissipation of fluid accumulation.

2) Moderate

Figure 4.5 (a) reveals a T2-FLAIR MRI image with multifocal, patchy hyperintensities within the bifrontal and right occipital subcortical white matter. Each lesion individually measures less than 5 cm, classifying them as mild; however, the presence of multiple lesions qualifies this case as moderate edema. The lesions demonstrate partial cortical involvement and mild local mass effect with subtle gyral swelling, consistent with clinically observed vasogenic edema. **Figure 4.5 (b)**, showing the synthetic lesion model, reproduces the most crucial features of moderate ARIA-E presentations, accurately reflecting multifocal hyperintensity distribution across frontal and occipital regions.





(a) MRI Scan Presenting Moderate Edema [19]. **(b)** Model of Moderate Edema.

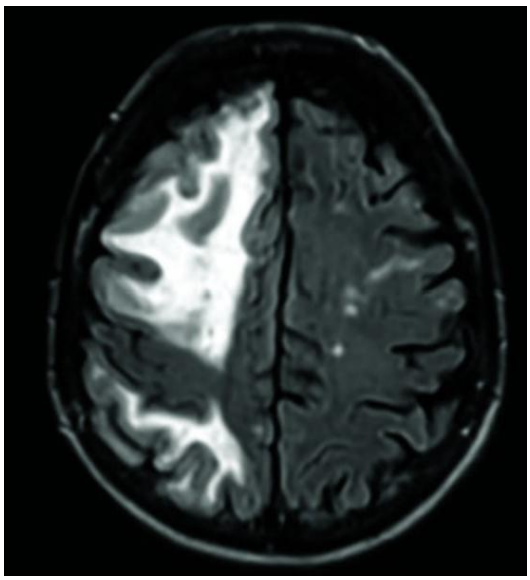
Figure 4.5- MRI vs Generated Model of Moderate Edema.

Gaussian distributions used in the model successfully produce realistic diffusion patterns consistent with fluid accumulation typically seen in clinical scenarios. Anatomical placement aligns closely with MRI findings, ensuring lesions predominantly occupy the white matter regions with each lesion being represented by a pixel value that matches the intensity of the MRI findings in each respective location.

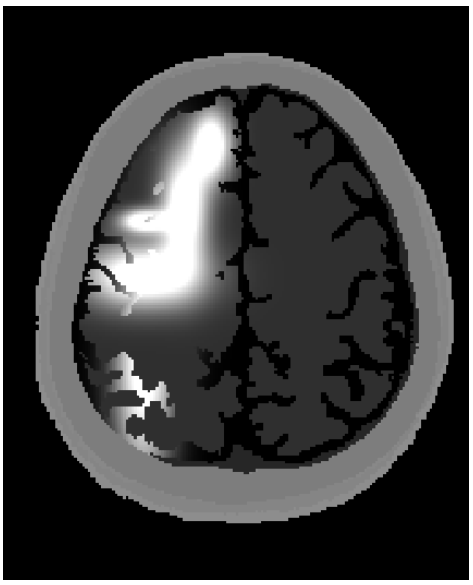
However, these intensity gradients within the synthetic lesions appear more uniform compared to the varying and irregular intensities observed clinically, resulting in a minor simplification of edema complexity. To enhance model accuracy, future iterations should incorporate adaptive intensity distributions with varying gradients as well as the rotation of the lesions at different angles within the DHP to better represent the irregular nature of the fluid accumulation and mild structural deformation evident in moderate edema lesions.

3) Severe

Figure 4.6 (a) demonstrates a post-treatment MRI showing extensive hyperintensity spanning large portions of the right frontal and parietal lobes, exceeding 10 cm in maximal dimension. This severe edema lesion is accompanied by substantial mass displacement and marked sulcal narrowing throughout the affected hemisphere, indicative of significant structural deformation. The synthetic severe ARIA-E model, illustrated by **Figure 4.6 (b),** displays a widespread and intense lesion distribution predominantly occupying the right frontoparietal region, closely matching the anatomical patterns observed clinically.



(a) MRI Scan Presenting Severe Edema [19].



(b) Model of Severe Edema.

Figure 4.6- MRI vs Generated Model of Severe Edema.

Gaussian intensity profiles produce realistic gradients, capturing the central intensity and gradual peripheral attenuation typical of severe fluid accumulation. However, notable differences emerge when comparing the model with the clinical image. Specifically, the model lacks representation of the pronounced narrowing in the sulci and random displacement evident in the MRI due to the limitations of static phantom anatomy. Additionally, the generated lesion portrays a smoother, more uniform distribution compared to the irregular intensities commonly encountered in extensive vasogenic edema. Incorporating structural distortions, including sulcal effacement and variable intensity transitions, would yield a more accurate simulation of severe edema pathology.

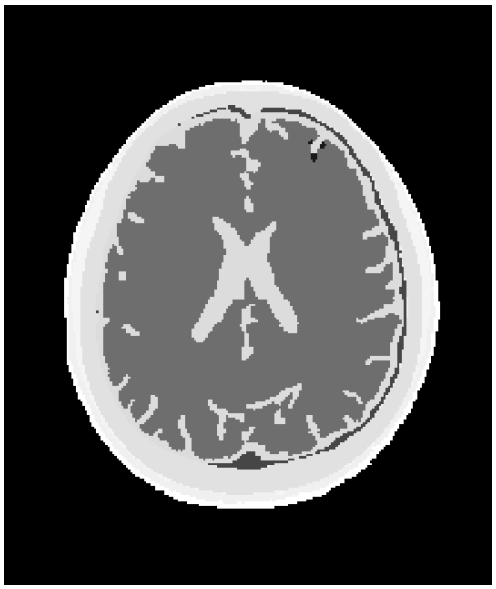
4.2.3 Microhaemorrhages

1) Mild

Figure 4.7 (a) demonstrates a post-treatment MRI presenting a limited number (<5) of small, discrete hypointense lesions in the peripheral left frontal region, consistent with mild microhaemorrhages. The subtle and isolated nature of these lesions, combined with the absence of significant mass displacement or extensive cortical involvement, clearly supports a mild severity classification. The synthetic model displayed in **Figure 4.7 (b)** places three punctate hypointensities strategically positioned in the frontal subcortical region, capturing the typical appearance of mild ARIA-H microhaemorrhages.



(a) MRI Scan Presenting Mild Microhaemorrhage [19].



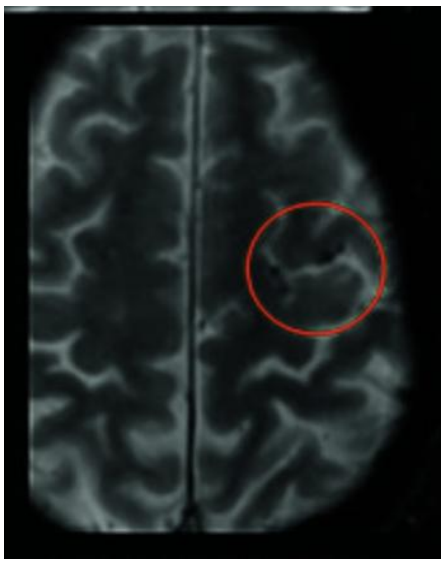
(b) Model of Mild Microhaemorrhage.

Figure 4.7- MRI vs Generated Model of Mild Microhaemorrhage.

While the distribution and localisation closely resemble clinical observations, the lesions appear uniform, lacking the nuanced irregularity observed in real cases. Improving the model could involve introducing variability in lesion shape and incorporating a mild, diffuse surrounding hyperintensity to better emulate the complex interplay of blood products and associated edema characteristic of mild ARIA-H.

2) Moderate

The postdosing GRE MRI in **Figure 4.8 (a)** reveals five distinct, treatment-emergent hypointense microhaemorrhages clustered within the left frontal region, indicative of moderate ARIA-H. These lesions appear as punctate dark spots, reflecting small deposits of blood products within the cerebral tissue. Their multiplicity classifies this presentation as moderate, highlighting increased vascular compromise compared to mild, isolated lesions. The synthetic model replicates the clustering of moderate ARIA-H lesions, depicting multiple discrete hypointensities accurately localised within the left frontal region. **Figure 4.8 (b)** presents the generated model depicting multiple discrete hypointensities accurately localised within the left frontal region.



(a) MRI Scan Presenting Moderate Microhaemorrhage [19].



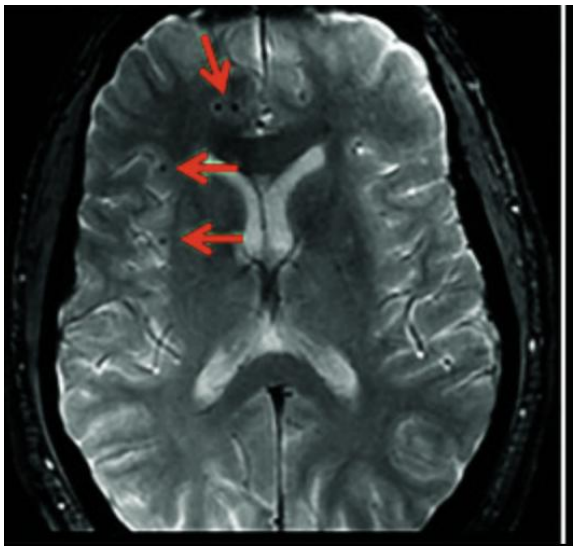
(b) Model of Moderate Microhaemorrhage.

Figure 4.8- MRI vs Generated Model of Moderate Microhaemorrhage.

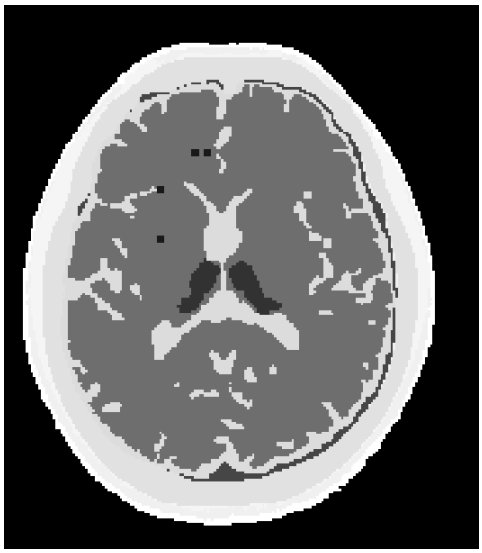
The model appropriately positions these lesions within cortical and subcortical structures, capturing the typical appearance and distribution observed clinically. However, while anatomically precise, the model demonstrates simplified lesion morphologies, lacking the subtle irregularities that are characteristic of real GRE imaging due to varying haemoglobin breakdown stages and local tissue reactions. Future refinement could benefit from incorporating more irregular lesion boundaries and intensity variations to achieve more authentic representation of moderate ARIA-H pathology.

3) Severe

Figure 4.9 (a) demonstrates multiple scattered treatment emergent hypointense lesions indicative of severe ARIA-H in an MRI scan. These microhaemorrhages are distributed within the cerebral tissue rather than clustered, highlighting significant widespread vascular compromise. Due to the extensive and scattered nature of these lesions, current clinical guidelines recommend discontinuation of anti-amyloid therapy. The synthetic model in **Figure 4.9 (b)** replicates the severe ARIA-H pathology by positioning multiple hypointense lesions throughout various cerebral regions.



(a) MRI Scan Presenting Severe Microhaemorrhage [19].



(b) Model of Severe Microhaemorrhage.

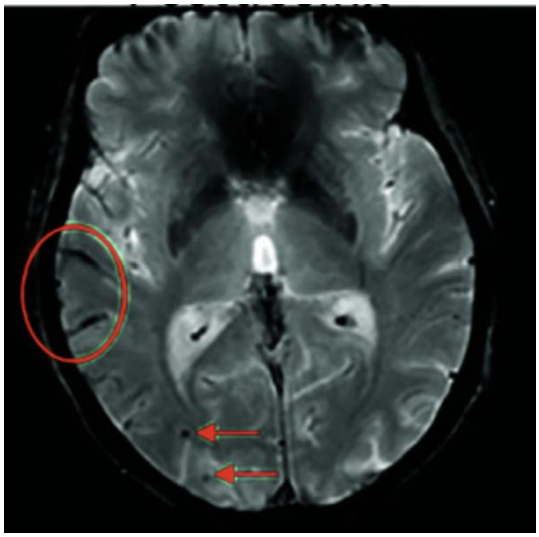
Figure 4.9- MRI vs Generated Model of Severe Microhaemorrhage.

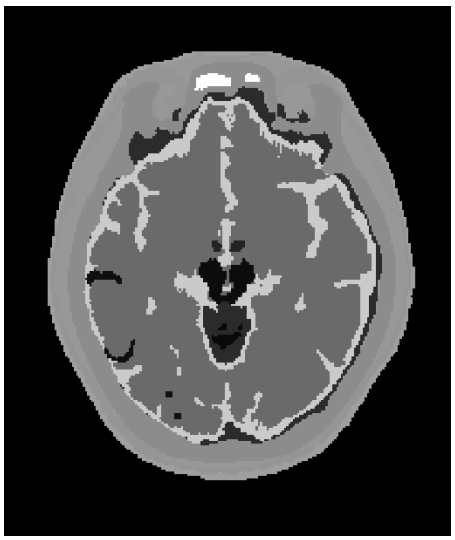
The scattered arrangement aligns appropriately with clinical observations, capturing the diffuse spatial distribution typical of severe ARIA-H. A critical evaluation identifies limitations in the uniformity of lesion shapes within the synthetic representation, whereas actual microhaemorrhages typically exhibit heterogeneous signal characteristics resulting from varied haemoglobin degradation stages. Enhancing future iterations with irregular and more spherical lesion shapes and variable intensity distributions would improve the clinical realism and diagnostic value of the generated ARIA-H models.

4.2.4 Superficial Siderosis

1) Mild

The postdosing GRE MRI in **Figure 4.10 (a)** demonstrates mild ARIA-H characterised by new superficial siderosis along contiguous sulci within the right temporal region. Additionally, two discrete treatment-emergent microhaemorrhages are identified in the right occipital lobe. The superficial siderosis manifests as linear hypointense regions along sulcal surfaces, while the microhaemorrhages appear as small, isolated hypointensities. Both findings align with the classification of mild ARIA-H, reflecting limited and localised haemorrhagic involvement.





(a) MRI Scan Presenting Mild Siderosis [19]**.** **(b)** Model of Mild Siderosis.

Figure 4.10- MRI vs Generated Model of Mild Siderosis.

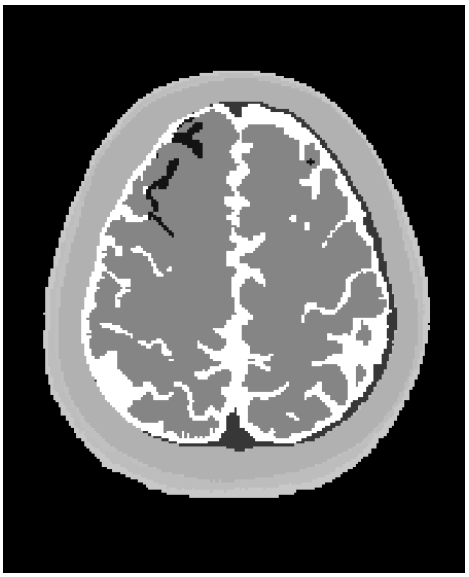
The synthetic model in **Figure 4.10 (b)** captures the presence of linear hypointensities corresponding to superficial siderosis in the right temporal region, closely reflecting the anatomical location and morphology observed clinically. The representation of siderosis as contiguous sulcal involvement provides a reasonable approximation of the actual pathology. Furthermore, the model accurately includes small isolated microhaemorrhages within the occipital lobe. However, limitations are noted in the uniform thickness and simplified linear appearance of siderosis, which contrasts with the irregular contours and variable widths typically observed in clinical GRE imaging. This is due to the synthetic siderosis model being restricted to the cerebral cortex gaps available in this specific slice of the DHP. Similarly, microhaemorrhages exhibit overly uniform and square appearances due to pixel-level constraints. These factors slightly limit the biological realism of the generated lesions.

2) Moderate

The GRE MRI acquired post-treatment, shown in **Figure 4.11 (a),** reveals two distinct regions of treatment-emergent superficial siderosis located bilaterally within the frontal lobes, with the right hemisphere showing more extensive involvement compared to the left. These siderotic lesions manifest as linear, sulcal-based hypointensities, reflective of hemosiderin deposition along cerebral sulci. The bilateral distribution and presence of multiple separate foci along with a punctuate microhaemorrhage signal located in the left region, classify this presentation as moderate ARIA-H.



(a) MRI Scan Presenting Moderate Superficial Siderosis [19].



(b) Model of Moderate Superficial Siderosis.

Figure 4.11- MRI vs Generated Model of Moderate Superficial Siderosis.

The synthetic ARIA-H model depicts two separate regions of superficial siderosis, as seen in **Figure 4.11 (b),** with a more prominent representation in the right frontal lobe, consistent with the clinical MRI findings. The linear, sulcal-aligned configuration of these lesions realistically simulates the anatomical presentation of siderosis. Nonetheless, the generated siderosis lesions appear thicker and less distributed across the right frontal lobe, due to the cortical constraints introduced by the DHP, although this specific slice was deemed the most suitable for the synthetic model. The singular microhaemorrhage signal is accurately represented in the left frontal lobe region, however it could be improved with an addition of a blurring effect.

4.3 Efficiency Analysis

4.3.1 Load DHP Function

Due to its significantly higher computational demands, the loading of the DHP data was evaluated and presented independently, **Figure 4.12** shows the mean execution time of the function over 20 iterations.

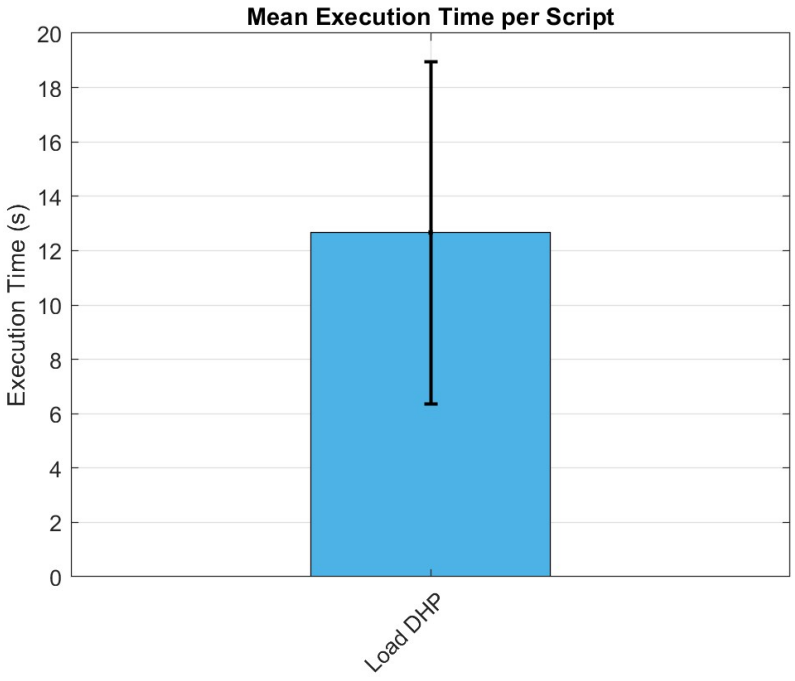


Figure 4.12- Mean Execution Time of the Load DHP Function.

The DHP loading script exhibited a mean execution time of approximately 12.7 seconds, with a standard deviation of ±6.3 seconds. Such variability is attributed primarily to input/output (I/O) operations involving the large three-dimensional dataset, system resource allocation, and background processes influencing the loading time. Memory profiling using MATLAB indicated a consistent peak memory usage of approximately 302 MB, reflecting the extensive nature of the dataset required for subsequent lesion generation processes.

The computational burden observed during the loading process highlights MATLAB's overhead inefficiencies in handling large volumetric datasets. The observed execution time is attributed to header parsing, memory reallocation, and autoscaling mechanisms, which introduce unnecessary delays compared to more direct binary file operations in lower-level programming environments. The significant discrepancy between the DHP loading time and the execution time of lesion generation scripts demonstrates that the primary computational limitation in this study stems from data handling rather than lesion modelling itself.

4.3.2 Lesion Generation Functions

The lesion generation scripts, when evaluated independently from the DHP loading phase, demonstrated significantly lower execution times. As illustrated in the collective execution time

graph, lesion generation scripts consistently displayed mean execution times between 0.68 seconds and 1.83 seconds, with standard deviations ranging from ±0.1 to ±0.2 seconds.

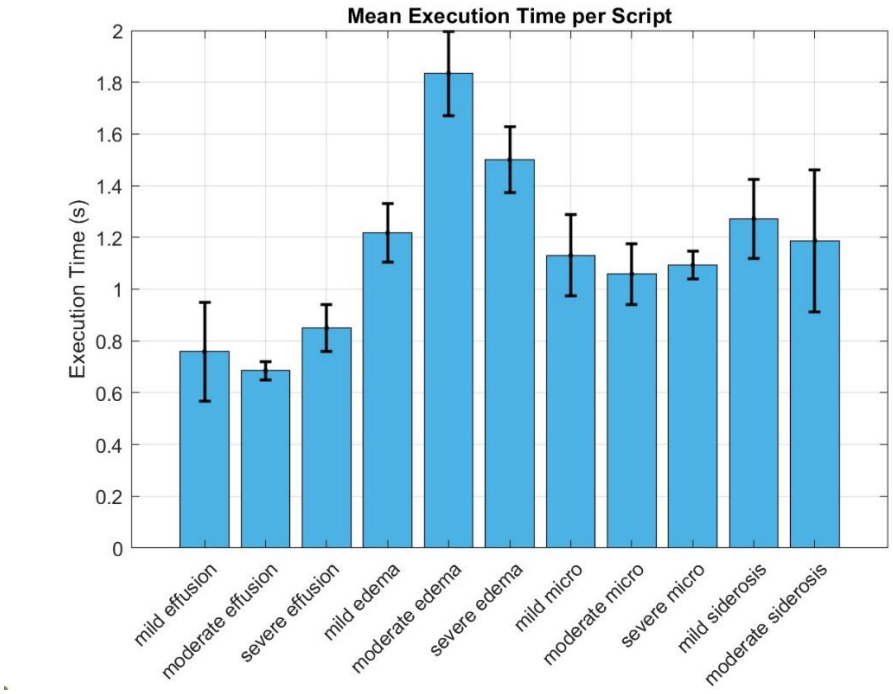


Figure 4.13- Mean Execution Time of the Lesion Generation Functions.

Scripts generating more complex lesions, such as moderate edema, displayed slightly higher execution times (1.83 ± 0.1 seconds), whereas simpler lesion models, such as moderate effusion, were executed more swiftly (0.68 ± 0.1 seconds). These minor variations confirm the computational efficiency of lesion generation, highlighting the algorithms' scalability and practical applicability to research scenarios demanding rapid and repetitive lesion synthesis.

Peak memory measurements across lesion generation scripts demonstrated a maximum incremental usage of approximately 4 KB over the baseline 302 MB used by the DHP. This negligible incremental memory confirms that memory usage by individual lesion generation scripts is minimal, suggesting highly efficient computations and minimal temporary variable allocation. This confirms that the primary computational burden lies in the initial data loading rather than the lesion generation process itself, reinforcing the need for optimised data handling strategies.

4.3.3 Feasibility

The findings confirm that the lesion generation methodology is computationally feasible and scalable for generating synthetic datasets for AI training. The efficiency of the lesion generation phase ensures that the approach remains viable for large scale data production. However, the

performance limitations observed in data loading highlight potential areas for future optimisation, particularly in selecting more efficient programming environments suited to handling large medical imaging datasets.

4.4 Effectiveness and Impact

This project contributes a scalable and anatomically grounded solution to a persistent challenge in computational neuroimaging: the limited availability of clinically annotated data for Amyloid-Related Imaging Abnormalities. By establishing a reproducible framework for generating biologically realistic ARIA-E and ARIA-H lesions within the Digital Human Phantom, this work enables the creation of large, labelled datasets without reliance on real patient scans. This significantly reduces the dependency on rare ARIA-positive MRI cases, which remain scarce in both clinical and research settings due to the low incidence rate and ethical constraints on data sharing.

The generated volumes, exported as sequential '.pgm' files, offer a structured and accessible format that can be directly utilised by scientists and engineers working in algorithm development. Each voxel in the exported dataset retains a consistent, class-based intensity value, allowing for seamless assignment of physical or diagnostic properties. This makes the synthetic DHP not only suitable for machine learning model training, but also compatible with voxel-based simulation frameworks for EM wave propagation studies. Researchers can assign tissue-specific dielectric values to each voxel, input the volumes into FDTD solvers, and analyse how EM fields interact with pathological brain regions. In this way, the same dataset supports both forward simulation and inverse problem-solving pipelines.

Beyond ARIA detection, this modelling approach presents a cost effective and ethically favourable strategy for dataset generation in neuroimaging and biomedical simulation research. It allows for rapid, repeatable testing of signal-processing algorithms, image reconstruction models, and diagnostic classifiers, particularly in early-stage research where access to real clinical data is restricted.

However, the current implementation is not without limitations. MATLAB's computational inefficiencies—particularly in memory allocation and file I/O—present challenges when scaling to high-volume datasets. The lack of quantitative validation against clinical images also limits the ability to benchmark morphological realism. These challenges highlight several promising

directions for future work, particularly in enhancing physical accuracy, automating lesion generation, and improving compatibility with high-performance computing environments.

4.5 Future Works

Building on the synthetic modelling framework established in this project, future research should prioritise refinements that enhance both the realism and utility of the generated datasets. While the current approach successfully models ARIA-E and ARIA-H within a biologically plausible anatomical scaffold, there is clear potential to increase the clinical fidelity, computational performance, and cross-domain applicability of the system.

The immediate priority for future research is refining the biological realism and accuracy of ARIA-H models. Currently, ARIA-E representations exhibit a higher degree of morphological and anatomical fidelity compared to ARIA-H models. Future works should therefore prioritise detailed improvements to microhaemorrhages and superficial siderosis representations by incorporating more realistic lesion boundaries and accurate spatial distributions to better mirror clinical MRI presentations.

Quantitative validation of generated models should also be a priority in future research. Implementing robust image similarity metrics, such as the Structural Similarity Index Measure (SSIM), would provide objective and numerical assessments of lesion model accuracy compared to clinical MRI images. SSIM offers a comprehensive assessment by accounting for luminance, contrast, and structural information, making it particularly suited for evaluating the quality of medical imaging models [39]. Integrating SSIM into routine validation practices would significantly enhance model credibility and provide clear benchmarks for ongoing methodological refinements.

Additionally, the project's MATLAB based implementation presents scalability challenges. The manual file handling, autoscaling behaviour, and memory allocation inefficiencies currently limit the ability to process large-scale datasets efficiently. Future iterations should explore transitioning to optimised environments—such as Fortran, Python with 'NumPy' or C—as well as implementing data streaming, parallelisation, or compression techniques to facilitate real-time or high-throughput processing.

Therefore, focusing future research on refining ARIA-H modelling accuracy, integrating robust numerical validation measures such as SSIM and optimising computational efficiency will

significantly enhance the utility, applicability, and credibility of synthetic lesion generation in clinical research and artificial intelligence training frameworks.

5 Conclusions

This dissertation set out to address a critical barrier in the development of AI-assisted diagnostics for Alzheimer's treatment: the limited availability of Amyloid-Related Imaging Abnormality data for algorithm training. To overcome this, the project developed a comprehensive pipeline for generating anatomically accurate, synthetic ARIA lesions—encompassing both ARIA-E and ARIA-H—within a DHP. The resulting models were exported in '.pgm' format to ensure compatibility with EM simulation environments and AI frameworks.

Through the integration of mathematically defined Gaussian and binary threshold models, the project accurately recreated lesion morphology and spatial distribution in accordance with clinical MRI features. White matter and cortical masking techniques were implemented to restrict lesions to biologically plausible tissue domains. This methodological fidelity was consistently reflected in the results, which demonstrated a close visual correspondence between synthetic lesions and real clinical examples across varying severity levels.

Critically, the generated models offer more than visual realism—they present a practical and scalable solution for the creation of large synthetic datasets. The '.pgm' files retain voxel-level precision, enabling their direct application in algorithm training pipelines and subsequent use in EM wave propagation studies.

While the primary aim of generating high-quality ARIA models was successfully met, the study also illuminated key areas for continued development. Notably, ARIA-H lesions remain more challenging to replicate with full morphological realism, and future validation using similarity metrics such as SSIM is necessary to quantitatively benchmark anatomical fidelity. Furthermore, the computational inefficiencies associated with MATLAB underscore the importance of transitioning to more optimised environments for future large-scale implementations.

Overall, this work provides a robust foundation for the synthetic modelling of ARIA, bridging a gap between medical imaging, algorithm development, and EM simulation. It not only addresses immediate training data shortages but also opens new avenues for interdisciplinary research that could enhance diagnostic capabilities and treatment planning in neurodegenerative disease.

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Appendices

A Project outline

Motivation and Aim:

Alzheimer's disease treatment can slow its progression at the stage of subjective cognitive decline or early mild cognitive impairment using emerging therapies like Lecanemab. But these therapies may cause complications such as amyloid related imaging abnormalities (ARIA-Edema/Haemorrhage). Regular MRI monitoring is necessary, but this is time and cost consuming as the scans must be analysed by a healthcare expert. Automating this process using AI can significantly improve efficiency. However, the lack of MRI images featuring ARIA limits AI training.

This project aims to examine currently available MRI images with ARIA, understand their features and translate them mathematically, and develop models of ARIA in the digital human phantom (DHP) provided by RIKEN using MATLAB. These models will generate synthetic MRI data to support future AI training for automatic ARIA detection.

Time Plan:

Semester 1:

- **S1W1-S1W3:**
    - Collect ARIA-E and ARIA-H images from the internet and keep links to the sources.
    - Conduct a literature review on ARIA, DHP, and image data structure.
    - Submit the project outline, plan, and risk assessment to Blackboard.
- **S1W4-S1W5:**
    - Obtain feedback and agree on the final project plan.

- Work on the extraction of a single plane (x, y, or z) from the DHP as initial practice in MATLAB.

- **S1W6-S1W10:**
    - Develop a MATLAB script to extract 2D slices from DHP data in PGM format.
    - Manipulate these slices to simulate ARIA lesions numerically, using PGM colour code mappings.
    - Implement the ARIA-E features before ARIA-H, focusing on lesion creation using the unused PGM values for human tissues.
- **S1W10-S1W11:**
    - Present progress on 2D manipulation and lesion creation.
    - Submit the draft introduction of the report to Blackboard.

Semester 2:

- **S2W1-S2W3:**
    - Develop methods to extract and manipulate 3D volumes from the DHP.
    - Focus on simulating more complex 3D ARIA lesions.
- **S2W4-S2W7:**
    - Develop numerical methods for comparing the simulated ARIA lesions with actual MRI images.
    - Submit a draft of the project report.
    - Present final stages of the project, focusing on accuracy validation and efficiency of the simulation code.
- **S2W8-S2W10:**
    - Finalise the report, prepare the presentation, and submit readme of the code.

B Risk Register

MANCHESTER 1824
The University of Manchester

RISK REGISTER FOR 3rd YEAR PROJECT

Project Title:	Introduction of the amyloid-related imaging abnormalities edema (ARIA-E) and hemorrhage (ARIA-H) to the brain.		Submission Date:	10/10/2024	
Student Name:	Qais	Momani			

Project Risk	Severity			Potential			Score (Severity x Potential)	Mitigation Measures
	L	M	H	L	M	H	L=1, M=2, H=3	
Eye strain due to prolonged periods in contact with screen		✓			✓		4	Ensure adequate lighting to reduce glare. Use screen filters or adjust display brightness.
back, neck, and shoulder pain.		✓			✓		4	Take regular breaks to stretch. Use an ergonomic chair and ensure proper desk setup.
Repetitive Strain Injury (RSI)			✓		✓		6	Take breaks to stretch fingers and wrists.
Difficulty in developing algorithms in MATLAB for ARIA-E/ARIA-H		✓		✓			2	Practice basic MATLAB programming, refer to coding resources, ask for Fortran code help from supervisor if stuck
Failure to meet deadlines			✓	✓			3	Set clear weekly goals, review time plan and adjust regularly, flag issues early with Fumie.

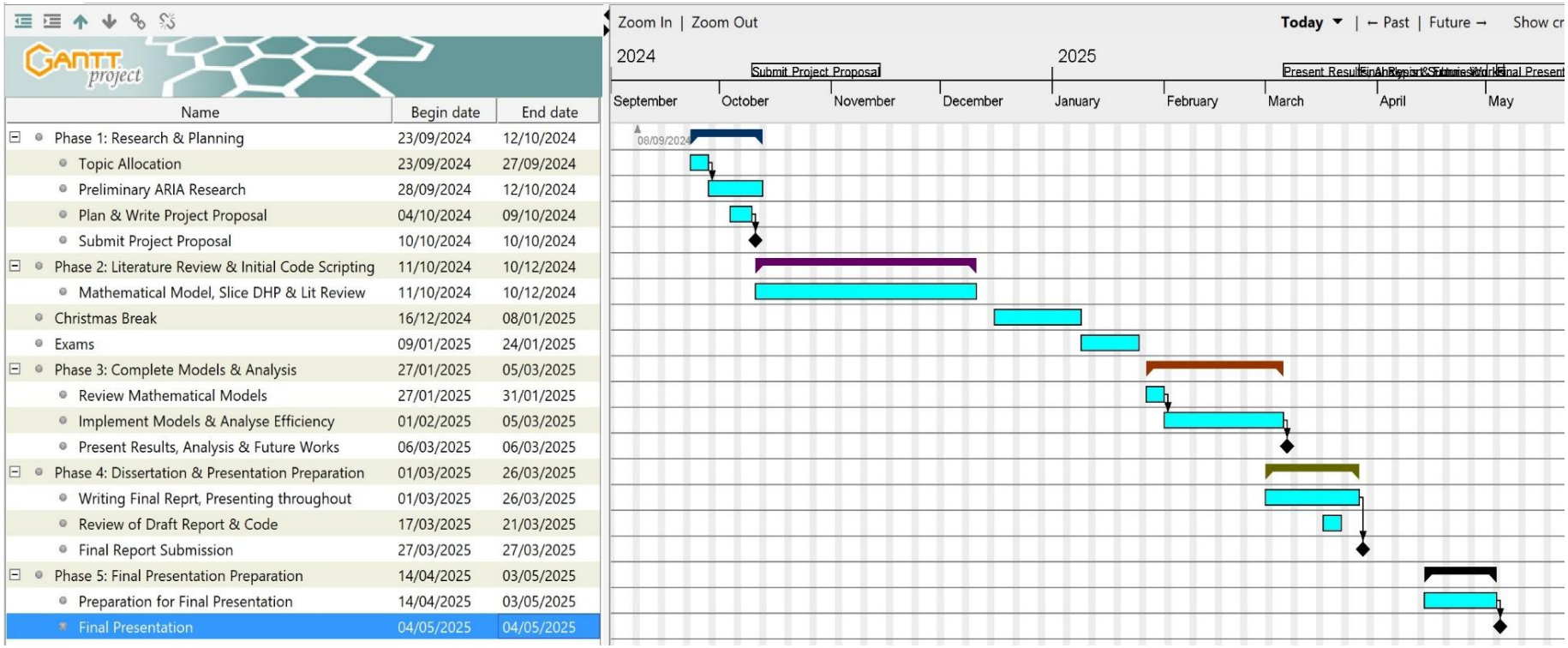
C H&S Risk Assessment

General Risk Assessment Form

Date: 4/04/2025	Assessed by: University of Manchester	Checked / Validated* by:	Location: Home University Computer Cluster	Assessment ref no	Review date: 11/04/2025
Task / premises: Working from home/ Working from university computer clusters For the latest information on coronavirus please visit the StaffNet website					

Activity	Hazard	Who might be harmed and how	Existing measures to control risk	Risk rating	Result
Working from home	Lone working	Student Isolated	1. Consulting university well-being services 2. Keeping regular contact with project supervisor	Low	A
Working from home / computer cluster	Stress / Wellbeing	Student Anxiety	1. Clearly set working hours and desired deadlines. 2. Open up about feelings of stress and anxiety to supervisor or colleagues.	Low	A
Use of electrical appliances	Misuse of electrical appliance, faulted electrical appliance.	Student Electric shock, burns and fire	1. Visual checks before use to make sure equipment, cables and free from defects 2. Liquid spills cleaned up immediately 3. Defective plugs, cables and equipment should be taken out of use	Med	A

D Gantt Chart



Name	Begin date	End date
Phase 1: Research & Planning	23/09/2024	12/10/2024
Topic Allocation	23/09/2024	27/09/2024
Preliminary ARIA Research	28/09/2024	12/10/2024
Plan & Write Project Proposal	04/10/2024	09/10/2024
Submit Project Proposal	10/10/2024	10/10/2024
Phase 2: Literature Review & Initial Code Scripting	11/10/2024	10/12/2024
Mathematical Model, Slice DHP & Lit Review	11/10/2024	10/12/2024
Christmas Break	16/12/2024	08/01/2025
Exams	09/01/2025	24/01/2025
Phase 3: Complete Models & Analysis	27/01/2025	05/03/2025
Review Mathematical Models	27/01/2025	31/01/2025
Implement Models & Analyse Efficiency	01/02/2025	05/03/2025
Present Results, Analysis & Future Works	06/03/2025	06/03/2025
Phase 4: Dissertation & Presentation Preparation	01/03/2025	26/03/2025
Writing Final Reprt, Presenting throughout	01/03/2025	26/03/2025
Review of Draft Report & Code	17/03/2025	21/03/2025
Final Report Submission	27/03/2025	27/03/2025
Phase 5: Final Presentation Preparation	14/04/2025	03/05/2025
Preparation for Final Presentation	14/04/2025	03/05/2025
Final Presentation	04/05/2025	04/05/2025

E Continuing Professional Development Log

Name: *Qais Momani*

Current and recent CPD activity:

CPD Activity Title	Description	Dates	CPD Hours
MATLAB Advanced Scripting	Developed complex MATLAB scripts for modelling ARIA lesions, using Gaussian and binary segmentation logic.	Oct 2024 – Feb 2025	30 hrs
Literature Review and Referencing	Conducted in-depth review of clinical and imaging literature on ARIA, MRI, PGM Data and Brain Anatomy, compiling 39 references.	Sept – Nov 2024	20 hrs
Image Processing and PGM Handling	Learned and applied image preprocessing, segmentation and export in PGM format for EM simulation usage.	Jan – Mar 2025	18 hrs
Technical Writing	Authored a 35-page dissertation and applied IEEE referencing style consistently. Refined dissertation structure based on feedback.	Feb – Apr 2025	15 hrs

Planned and Future CPD activity:

CPD Activity Title	Description	Skills addressed	Dates
Research Presentation and Communication	Prepare and deliver an academic poster or presentation summarising key findings.	Academic communication, clarity, audience engagement	May 2025
Advanced MATLAB and Data Handling	Further develop MATLAB proficiency with focus on 3D matrix operations, image processing and automation.	Improved code optimisation, data management, and computational efficiency	Apr – Sep 2025
Algorithm Training for Biomedical Data	Gain experience in applying PGM image data in algorithm training for pattern recognition and simulations.	Machine learning integration, data labelling, evaluation metrics	Jun - Sep 2025

F Use of AI Assistance

During the completion of this project and preparation of this report, AI-based tools such as ChatGPT (OpenAI) were used to support code optimisation, text refinement, and language clarity. All generated content was critically reviewed and edited to ensure factual accuracy, consistency with source materials, and academic integrity.

G Magnetic Resonance Imaging

Magnetic Resonance Imaging (MRI) is a non-invasive medical imaging technique that utilises strong magnetic fields, radiofrequency pulses, and computer processing to generate detailed images of internal body structures. Unlike X-rays or computed tomography (CT), MRI does not use ionising radiation, making it suitable for repeated imaging, particularly beneficial for longitudinal monitoring of conditions such as Alzheimer's disease and its associated abnormalities [37].

MRI provides high-resolution images of various tissues, including brain structures, and is extensively utilised for neurological assessments, tumour detection, vascular abnormalities, inflammatory conditions, and degenerative diseases. Its unique ability to differentiate between soft tissues, detect subtle pathological changes, and identify early disease markers makes MRI invaluable in both clinical and research settings [37].

Common types of MRI scans include:

T1-weighted imaging: Excellent anatomical detail, used widely for brain structure assessment.

T2-weighted imaging: Highlights fluid and inflammation, essential for detecting edema.

FLAIR (Fluid-Attenuated Inversion Recovery): Suppresses fluid signals to improve visualisation of lesions near cerebrospinal fluid (CSF).

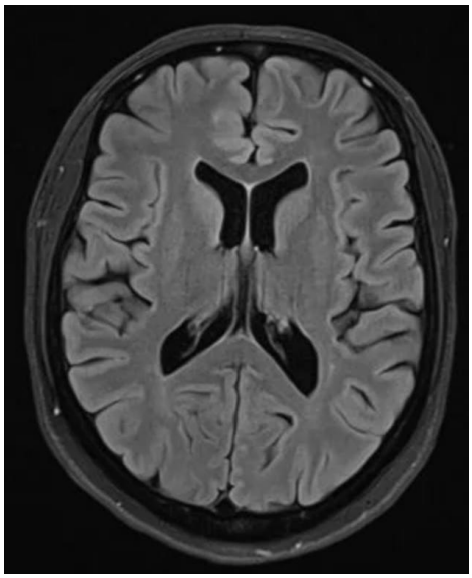
T2*-weighted Gradient-Recalled Echo (GRE): Sensitive to magnetic susceptibility, ideal for detecting microhaemorrhages.

Diffusion-weighted imaging (DWI): Evaluates tissue integrity based on water diffusion, critical for stroke assessment.

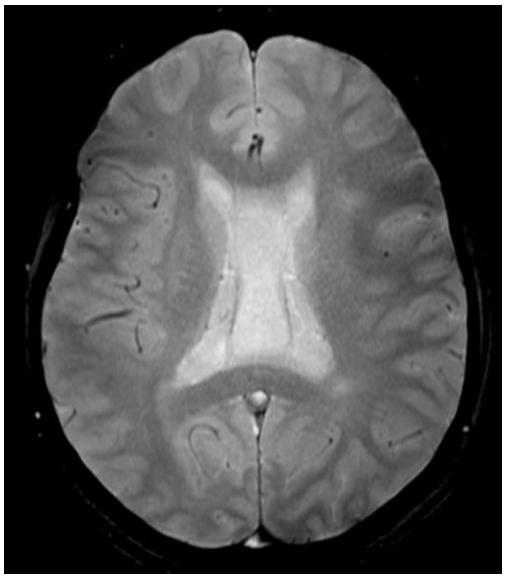
Susceptibility-weighted imaging (SWI): Enhanced sensitivity to blood products and microhaemorrhages, complementary to T2*-GRE [38].

For the detection and analysis of ARIA subtypes in Alzheimer's disease, two MRI sequences are primarily employed:

- **T2-FLAIR** for ARIA-E: Effectively visualises vasogenic edema and sulcal effusion, presenting as hyperintense regions around the cortical sulci and within brain parenchyma, shown in **Figure 5.1 (a)**.
- ***T2-GRE**** for ARIA-H: Sensitive to haemorrhagic lesions including cerebral microhaemorrhages and superficial siderosis, appearing as hypointense signals due to iron deposition from haemoglobin breakdown products, shown in **Figure 5.1 (b)** [38].



(a) T2 FLAIR MRI Image [37].



(b) T2-GRE* MRI Image [38].

Figure 5.1- T2 FLAIR and T2*-GRE MRI images for Visualisation.

This project specifically utilises these two MRI sequences, aligning with standard clinical practice for ARIA monitoring.

despite its substantial clinical benefits, incurs significant financial and operational costs globally. In the United Kingdom, the average cost of a single MRI examination ranges from £300 to £500 [32].

In the United States, MRI costs are notably higher, ranging between $1,000 and $3,500 per scan, influenced by healthcare system structures, insurance coverage variability, and higher operational

overheads. MRI expenditures constitute a substantial portion of healthcare budgets, estimated at several billion dollars annually [33].

Globally, access and costs vary significantly, with developed countries generally having widespread MRI availability at high operational expense, while developing regions face limited access due to prohibitive costs and infrastructure constraints. The high economic burden associated with MRI necessitates efficient and targeted utilisation, underscoring the importance of cost-effective imaging strategies, especially in chronic conditions requiring repeated imaging such as Alzheimer's disease and ARIA.

This financial and operational burden highlights the value of developing automated and efficient diagnostic tools to reduce overall costs and improve resource allocation.

H MATLAB Scripts Source Code

1. Load DHP

```
% Specify the directory where PGM files are stored
folderPath = 'C:\Uni_stuff\Individual_project\pgmdata\Extracted_head';

% Get a list of PGM files in the directory
files = dir(fullfile(folderPath, '*.pgm'));
numFiles = numel(files);

% Pre-allocate the 3D array based on the number of files and image size
% Assuming all images have the same size, use the first image to get dimensions
firstFilePath = fullfile(folderPath, files(1).name);
fid = fopen(firstFilePath, 'r');
fgetl(fid); % Skip the magic number line
fgetl(fid); % Skip the comment line
dims = sscanf(fgetl(fid), '%d %d'); % Read image dimensions
fgetl(fid); % Skip the max value line
fclose(fid);

width = dims(1);
height = dims(2);
DHP_3D = zeros(height, width, numFiles); % Pre-allocate 3D array

% Loop through each file, load image data into the 3D array
for i = 1:numFiles
    filePath = fullfile(folderPath, files(i).name);
    fid = fopen(filePath, 'r');

    % Skip header lines
    fgetl(fid); % Skip the magic number line
    fgetl(fid); % Skip the comment line
    fgetl(fid); % Skip width and height line
    fgetl(fid); % Skip the max value line

    % Read the image data
    data = fscanf(fid, '%d', [width, height]); % Read as a 2D array
    fclose(fid);

    % Insert into 3D array, note the transpose operation because fscanf fills
columns first
    DHP_3D(:, :, i) = data';
End
```

2. Mild Effusion

```
% Parameters for the model

sigma = 7; % Adjust based on the severity level
centerCoordinates = [100, 290, 95]; % Adjust these coordinates based on the target
area in the DHP

% Size of the 3D array
xSize = 265;
ySize = 490;
```

```
zSize = 305;


% Generate a 3D Gaussian blob at the specified location
[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);
effusionModel = exp(-((x-centerCoordinates(1)).^2 + (y-centerCoordinates(2)).^2 +
(z-centerCoordinates(3)).^2)/(2*sigma^2));

% Insert the effusion into the DHP
% Assuming effusion model intensity needs to be normalized or scaled according to
the DHP
peakIntensity = 60;
effusionModel = effusionModel * peakIntensity;

% Step 1: Identify White Matter in the Original DHP
whiteMatterMask = DHP_3D == 2;  % Binary mask where white matter exists
blackRegionMask = DHP_3D == 1;      % cortical regions with pixel value 1

% Step 2: Create a Copy of the White Matter with Higher Intensity for Visualization
enhancedWhiteMatter = whiteMatterMask * 17;  % Assign a brighter pixel value (17) to
the white matter

% Step 3: Overlay the Enhanced White Matter on Top of the Lesion Model
% Combine the lesion model with enhanced white matter (without modifying original
DHP)
DHP_eff_mild = DHP_3D;
DHP_eff_mild(blackRegionMask) = effusionModel(blackRegionMask);  % Apply lesions
only in black areas
DHP_eff_mild(whiteMatterMask) = enhancedWhiteMatter(whiteMatterMask);  % Overlay
brighter white matter


% Set the constant z value
zconstant = 95; % example value

% Check that zconstant is within the image dimensions
if zconstant > size(DHP_eff_mild, 3)
    error('The constant z value is out of the image dimension bounds.');
end

% Extract the plane at z = zconstant
extracted_plane_z = squeeze(DHP_eff_mild(:, :, zconstant));

% Display the extracted plane
figure;
imshow(extracted_plane_z, []);
title(sprintf('Extracted Plane at z = %d', zconstant));
```

3. Moderate Effusion

```
% Parameters for the model
sigmaX_1 = 13; % Adjust based on the severity level
sigmaY_1 = 15;
sigmaZ_1 = 2;
centerCoordinates = [172, 307, 82]; % Adjust these coordinates based on the target
area in the DHP

% Size of the 3D array
```

```
xSize = 265;
ySize = 490;
zSize = 305;


% Generate a 3D Gaussian blob at the specified location
[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);
effusionModel_1 = exp(-((x-centerCoordinates(1)).^2 / (2*sigmaX_1^2) + (y-
centerCoordinates(2)).^2 / (2*sigmaY_1^2) + (z-centerCoordinates(3)).^2 /
(2*sigmaZ_1^2)));


% Insert the effusion into the DHP
% Assuming effusion model intensity needs to be normalized or scaled according to
the DHP
peakIntensity = 60;
effusionModel_1 = effusionModel_1 * peakIntensity;



% Step 1: Identify White Matter and Black Regions in the Original DHP
whiteMatterMask = DHP_3D == 2;      % White matter with pixel value 2
blackRegionMask = DHP_3D == 1;      % Black regions with pixel value 1

% Step 2: Create a Copy of the White Matter with Higher Intensity for Visualization
enhancedWhiteMatter = whiteMatterMask * 17;  % Assign a brighter pixel value to the
white matter

% Step 3: Apply Lesions Only to Black Regions
DHP_eff_moderate = DHP_3D;  % Start with the original DHP
DHP_eff_moderate(blackRegionMask) = effusionModel_1(blackRegionMask);  % Apply
lesions only in black areas

% Step 4: Reapply the Enhanced White Matter on Top
DHP_eff_moderate(whiteMatterMask) = enhancedWhiteMatter(whiteMatterMask);  % Overlay
brighter white matter


% Set the constant z value
zconstant = 82; % example value

% Check that zconstant is within the image dimensions
if zconstant > size(DHP_eff_moderate, 3)
    error('The constant z value is out of the image dimension bounds.');
end

% Extract the plane at z = zconstant
extracted_plane_z = squeeze(DHP_eff_moderate(:, :, zconstant));

% Display the extracted plane
figure;
imshow(extracted_plane_z, []);
title(sprintf('Extracted Plane at z = %d', zconstant));
```

4. Severe Effusion

```
% Parameters for the model
```

```
sigmaX_1 = 10; % Adjust based on the severity level
sigmaY_1 = 2;
sigmaZ_1 = 3;
centerCoordinates = [125, 250, 82]; % Adjust these coordinates based on the target
area in the DHP

% Size of the 3D array
xSize = 265;
ySize = 490;
zSize = 305;


% Generate a 3D Gaussian blob at the specified location
[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);
effusionModel_1 = exp(-((x-centerCoordinates(1)).^2 / (2*sigmaX_1^2) + (y-
centerCoordinates(2)).^2 / (2*sigmaY_1^2) + (z-centerCoordinates(3)).^2 /
(2*sigmaZ_1^2)));


% Parameters for the model
sigmaX_2 = 7; % Adjust based on the severity level
sigmaY_2 = 7;
sigmaZ_2 = 1;
centerCoordinates = [191, 223, 82]; % Adjust these coordinates based on the target
area in the DHP


% Generate a 3D Gaussian blob at the specified location
[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);
effusionModel_2 = exp(-((x-centerCoordinates(1)).^2 / (2*sigmaX_2^2) + (y-
centerCoordinates(2)).^2 / (2*sigmaY_2^2) + (z-centerCoordinates(3)).^2 /
(2*sigmaZ_2^2)));




% Insert the effusion into the DHP
% Assuming effusion model intensity needs to be normalized or scaled according to
the DHP
peakIntensity1 = 60;
peakIntensity2 = 34;
effusionModel_1 = effusionModel_1 * peakIntensity1;
effusionModel_2 = effusionModel_2 * peakIntensity2;

% Step 1: Identify White Matter in the Original DHP
whiteMatterMask = DHP_3D == 2;  % Binary mask where white matter exists

% Step 2: Create a Copy of the White Matter with Higher Intensity for Visualization
enhancedWhiteMatter = whiteMatterMask * 17;  % Assign a brighter pixel value (17) to
the white matter

% Step 3: Overlay the Enhanced White Matter on Top of the Lesion Model
% Combine the lesion model with enhanced white matter (without modifying original
DHP)
DHP_eff_severe = DHP_3D + effusionModel_1 + effusionModel_2; % Original DHP + Lesion
Model
DHP_eff_severe(whiteMatterMask) = enhancedWhiteMatter(whiteMatterMask);  % Overlay
brighter white m
```

```
% Set the constant z value
zconstant = 82; % example value

% Check that zconstant is within the image dimensions
if zconstant > size(DHP_eff_severe, 3)
    error('The constant z value is out of the image dimension bounds.');
end

% Extract the plane at z = zconstant
extracted_plane_z = squeeze(DHP_eff_severe(:, :, zconstant));

% Display the extracted plane
figure;
imshow(extracted_plane_z, []);
title(sprintf('Extracted Plane at z = %d', zconstant));
```

5. Mild Edema

```
% Size of the 3D array
xSize = 265;
ySize = 490;
zSize = 305;

% Parameters for the model

sigmaX_1 = 2; % Adjust based on the severity level
sigmaY_1 = 3;
sigmaZ_1 = 1;
centerCoordinates_1 = [204, 235, 92]; % Adjust these coordinates based on the target
area in the DHP

% Generate a 3D Gaussian blob at the specified location
[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);
edemaModel_1 = exp(-((x-centerCoordinates_1(1)).^2 / (2*sigmaX_1^2) + (y-
centerCoordinates_1(2)).^2 / (2*sigmaY_1^2) + (z-centerCoordinates_1(3)).^2 /
(2*sigmaZ_1^2)));

% Parameters for the model

sigmaX_2 = 4; % Adjust based on the severity level
sigmaY_2 = 10;
sigmaZ_2 = 1;
centerCoordinates_2 = [204, 223, 92]; % Adjust these coordinates based on the target
area in the DHP

% Generate a 3D Gaussian blob at the specified location
[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);
edemaModel_2 = exp(-((x-centerCoordinates_2(1)).^2 / (2*sigmaX_2^2) + (y-
centerCoordinates_2(2)).^2 / (2*sigmaY_2^2) + (z-centerCoordinates_2(3)).^2 /
(2*sigmaZ_2^2)));

% Parameters for the model

sigmaX_3 = 5; % Adjust based on the severity level
sigmaY_3 = 6;
sigmaZ_3 = 1;
```

```
centerCoordinates_3 = [198, 217, 92]; % Adjust these coordinates based on the target
area in the DHP

% Generate a 3D Gaussian blob at the specified location
[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);
edemaModel_3 = exp(-((x-centerCoordinates_3(1)).^2 / (2*sigmaX_3^2) + (y-
centerCoordinates_3(2)).^2 / (2*sigmaY_3^2) + (z-centerCoordinates_3(3)).^2 /
(2*sigmaZ_3^2)));


% Insert the effusion into the DHP
% Assuming effusion model intensity needs to be normalized or scaled according to
the DHP
peakIntensity_1 = 44;
peakIntensity_2 = 34;
peakIntensity_3 = 22;
edemaModel_1 = edemaModel_1 * peakIntensity_1;
edemaModel_2 = edemaModel_2 * peakIntensity_2;
edemaModel_3 = edemaModel_3 * peakIntensity_3;
% Step 1: Identify White Matter in the Original DHP
whiteMatterMask = DHP_3D == 2;  % Binary mask where white matter exists

% Step 2: Create a Copy of the White Matter with Higher Intensity for Visualization
enhancedWhiteMatter = whiteMatterMask * 17;  % Assign a brighter pixel value (17) to
the white matter

% Step 3: Apply Lesions Only to White Matter
lesionMask = whiteMatterMask;  % Lesions will only be applied where white matter
exists
DHP_edema_mild = DHP_3D;  % Start with the original DHP

% Step 4: Reapply the Enhanced White Matter for Visualization
DHP_edema_mild(whiteMatterMask) = enhancedWhiteMatter(whiteMatterMask);  % Ensure
white matter remains visible

DHP_edema_mild(lesionMask) = DHP_edema_mild(lesionMask) + edemaModel_1(lesionMask) +
edemaModel_2(lesionMask) + edemaModel_3(lesionMask);  % Apply lesions to white
matter

% Set the constant z value
zconstant = 92; % example value

% Check that zconstant is within the image dimensions
if zconstant > size(DHP_edema_mild, 3)
    error('The constant z value is out of the image dimension bounds.');
end

% Extract the plane at z = zconstant
extracted_plane_z = squeeze(DHP_edema_mild(:, :, zconstant));

% Display the extracted plane
figure;
imshow(extracted_plane_z, []);
title(sprintf('Extracted Plane at z = %d', zconstant));
```

6. Moderate Edema

```
% Parameters for the model

sigmaX_1 = 2; % Adjust based on the severity level
sigmaY_1 = 1;
sigmaZ_1 = 2;
centerCoordinates1 = [190, 272, 77]; % Adjust these coordinates based on the target
area in the DHP

% Size of the 3D array
xSize = 265;
ySize = 490;
zSize = 305;


% Generate a 3D Gaussian blob at the specified location
[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);
edemaModel_1 = exp(-((x-centerCoordinates1(1)).^2 / (2*sigmaX_1^2) + (y-
centerCoordinates1(2)).^2 / (2*sigmaY_1^2) + (z-centerCoordinates1(3)).^2 /
(2*sigmaZ_1^2)));


% Parameters for the model

sigmaX_2 = 3; % Adjust based on the severity level
sigmaY_2 = 4;
sigmaZ_2 = 2;
centerCoordinates2 = [90, 269, 77]; % Adjust these coordinates based on the target area
in the DHP


% Generate a 3D Gaussian blob at the specified location
[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);
edemaModel_2 = exp(-((x-centerCoordinates2(1)).^2 / (2*sigmaX_2^2) + (y-
centerCoordinates2(2)).^2 / (2*sigmaY_2^2) + (z-centerCoordinates2(3)).^2 /
(2*sigmaZ_2^2)));

% Parameters for the model

sigmaX_3 = 2; % Adjust based on the severity level
sigmaY_3 = 4;
sigmaZ_3 = 2;
centerCoordinates3 = [81, 211, 77]; % Adjust these coordinates based on the target area
in the DHP


% Generate a 3D Gaussian blob at the specified location
[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);
edemaModel_3 = exp(-((x-centerCoordinates3(1)).^2 / (2*sigmaX_3^2) + (y-
centerCoordinates3(2)).^2 / (2*sigmaY_3^2) + (z-centerCoordinates3(3)).^2 /
(2*sigmaZ_3^2)));


sigmaX_4 = 4; % Adjust based on the severity level
sigmaY_4 = 3;
sigmaZ_4 = 3;
centerCoordinates4 = [85, 272, 77]; % Adjust these coordinates based on the target area
in the DHP


% Generate a 3D Gaussian blob at the specified location
```

```
[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);
edemaModel_4 = exp(-((x-centerCoordinates4(1)).^2 / (2*sigmaX_4^2) + (y-
centerCoordinates4(2)).^2 / (2*sigmaY_4^2) + (z-centerCoordinates4(3)).^2 /
(2*sigmaZ_4^2)));


% Parameters for the model

sigmaX_5 = 6; % Adjust based on the severity level
sigmaY_5 = 6;
sigmaZ_5 = 5;
centerCoordinates5 = [76, 212, 77]; % Adjust these coordinates based on the target area
in the DHP


% Generate a 3D Gaussian blob at the specified location
[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);
edemaModel_5 = exp(-((x-centerCoordinates5(1)).^2 / (2*sigmaX_5^2) + (y-
centerCoordinates5(2)).^2 / (2*sigmaY_5^2) + (z-centerCoordinates5(3)).^2 /
(2*sigmaZ_5^2)));

% Insert the effusion into the DHP
% Assuming effusion model intensity needs to be normalized or scaled according to the
DHP
peakIntensity1 = 60;
peakIntensity2 = 22;
peakIntensity3 = 50;
peakIntensity4 = 22;
edemaModel_1 = edemaModel_1 * peakIntensity1;
edemaModel_2 = edemaModel_2 * peakIntensity2;
edemaModel_3 = edemaModel_3 * peakIntensity3;
edemaModel_4 = edemaModel_4 * peakIntensity4;
edemaModel_5 = edemaModel_5 * peakIntensity4;

% Step 1: Identify White Matter in the Original DHP
whiteMatterMask = DHP_3D == 2;  % Binary mask where white matter exists

% Step 2: Create a Copy of the White Matter with Higher Intensity for Visualization
enhancedWhiteMatter = whiteMatterMask * 17;  % Assign a brighter pixel value (17) to
the white matter

% Step 3: Apply Lesions Only to White Matter
lesionMask = whiteMatterMask;  % Lesions will only be applied where white matter exists
DHP_edema_moderate = DHP_3D;  % Start with the original DHP

% Step 4: Reapply the Enhanced White Matter for Visualization
DHP_edema_moderate(whiteMatterMask) = enhancedWhiteMatter(whiteMatterMask);  % Ensure
white matter remains visible

DHP_edema_moderate(lesionMask) = DHP_edema_moderate(lesionMask) +
edemaModel_1(lesionMask) + edemaModel_2(lesionMask) + edemaModel_3(lesionMask) +
edemaModel_4(lesionMask) + edemaModel_5(lesionMask);  % Apply lesions to white matter




% Set the constant z value
zconstant = 77; % example value

% Check that zconstant is within the image dimensions
```

```
if zconstant > size(DHP_edema_moderate, 3)
    error('The constant z value is out of the image dimension bounds.');
end

% Extract the plane at z = zconstant
extracted_plane_z = squeeze(DHP_edema_moderate(:, :, zconstant));

% Display the extracted plane
figure;
imshow(extracted_plane_z, []);
title(sprintf('Extracted Plane at z = %d', zconstant));
```

7. Severe Edema

```
% Parameters for the model
sigmaX_1 = 22; % Extend along X-axis
sigmaY_1 = 8; % Moderate spread along Y-axis
sigmaZ_1 = 5;  % Thin spread along Z-axis
centerCoordinates1 = [117, 275, 57];

sigmaX_2 = 30;
sigmaY_2 = 8;
sigmaZ_2 = 5;
centerCoordinates2 = [165, 285, 57];

sigmaX_3 = 15;
sigmaY_3 = 6;
sigmaZ_3 = 5;
centerCoordinates3 = [105, 270, 57];

sigmaX_4 = 8;
sigmaY_4 = 15;
sigmaZ_4 = 5;
centerCoordinates4 = [130, 285, 57];

sigmaX_5 = 3;
sigmaY_5 = 6;
sigmaZ_5 = 5;
centerCoordinates5 = [110, 285, 57];

sigmaX_6 = 9;
sigmaY_6 = 5;
sigmaZ_6 = 5;
centerCoordinates6 = [82, 260, 57];


xSize = 265;
ySize = 490;
zSize = 305;

[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);
edemaModel_1 = exp(-((x-centerCoordinates1(1)).^2/(2*sigmaX_1^2) + (y-
centerCoordinates1(2)).^2/(2*sigmaY_1^2) + (z-
centerCoordinates1(3)).^2/(2*sigmaZ_1^2)));
edemaModel_2 = exp(-((x-centerCoordinates2(1)).^2/(2*sigmaX_2^2) + (y-
centerCoordinates2(2)).^2/(2*sigmaY_2^2) + (z-
centerCoordinates2(3)).^2/(2*sigmaZ_2^2)));
```

```
edemaModel_3 = exp(-((x-centerCoordinates3(1)).^2/(2*sigmaX_3^2) + (y-
centerCoordinates3(2)).^2/(2*sigmaY_3^2) + (z-
centerCoordinates3(3)).^2/(2*sigmaX_3^2)));
edemaModel_4 = exp(-((x-centerCoordinates4(1)).^2/(2*sigmaX_4^2) + (y-
centerCoordinates4(2)).^2/(2*sigmaY_4^2) + (z-
centerCoordinates4(3)).^2/(2*sigmaX_4^2)));
edemaModel_5 = exp(-((x-centerCoordinates5(1)).^2/(2*sigmaX_5^2) + (y-
centerCoordinates5(2)).^2/(2*sigmaY_5^2) + (z-
centerCoordinates5(3)).^2/(2*sigmaX_5^2)));
edemaModel_6 = exp(-((x-centerCoordinates6(1)).^2/(2*sigmaX_6^2) + (y-
centerCoordinates6(2)).^2/(2*sigmaY_6^2) + (z-
centerCoordinates6(3)).^2/(2*sigmaX_6^2)));


peakIntensity = 100;
edemaModel_1 = edemaModel_1 * peakIntensity;
edemaModel_2 = edemaModel_2 * peakIntensity;
edemaModel_3 = edemaModel_3 * peakIntensity;
edemaModel_4 = edemaModel_4 * peakIntensity;
edemaModel_5 = edemaModel_5 * peakIntensity;
edemaModel_6 = edemaModel_6 * peakIntensity;

whiteMatterMask = DHP_3D == 2;
blackRegionMask = DHP_3D == 1;      % Black regions with pixel value 1

enhancedWhiteMatter = whiteMatterMask * 17;

DHP_edema_severe = DHP_3D;

DHP_edema_severe(blackRegionMask) = edemaModel_1(blackRegionMask) +
edemaModel_2(blackRegionMask);


DHP_edema_severe(whiteMatterMask) = enhancedWhiteMatter(whiteMatterMask);

DHP_edema_severe(whiteMatterMask) = DHP_edema_severe(whiteMatterMask) +
edemaModel_3(whiteMatterMask) + edemaModel_4(whiteMatterMask) +
edemaModel_5(whiteMatterMask) + edemaModel_6(whiteMatterMask);

figure;
zconstant = 57;
extracted_plane_z = squeeze(DHP_edema_severe(:, :, zconstant));
imshow(extracted_plane_z, 'DisplayRange', [min(DHP_3D(:)), max(DHP_3D(:))]);
```

8. Mild Microhaemmorhages

```
% Define the DHP dimensions (Ensure this matches your DHP dataset)
xSize = 265;
ySize = 490;
zSize = 305;

% Initialize a 3D meshgrid
[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);

% Define hemorrhage lesion parameters (3 Lesions)
lesionCenters = [
    68, 218, 77;  % Lesion 1 (Example coordinates)
```

```
    73, 221, 77;  % Lesion 2
    67, 222, 77   % Lesion 3
];

% Elliptical lesion dimensions
sigmaX = [1, 2, 1];  % Adjusted to form elliptical lesions
sigmaY = [2, 1, 1];
sigmaZ = [1, 1, 1];

% Create an empty 3D lesion model
hemorrhageModel = zeros(ySize, xSize, zSize);

% Generate microhemorrhage lesions at specified locations
for i = 1:size(lesionCenters, 1)
    cx = lesionCenters(i,1);
    cy = lesionCenters(i,2);
    cz = lesionCenters(i,3);

    % Elliptical lesion mask
    lesionMask = exp(-((x-cx).^2 / (2*sigmaX(i)^2) + (y-cy).^2 / (2*sigmaY(i)^2) +
(z-cz).^2 / (2*sigmaZ(i)^2)));

    % Apply threshold to create a hard boundary (binary mask)
    lesionMask = lesionMask > 0.5;

    % Assign a low pixel intensity (hypointense appearance)
    hemorrhageModel(lesionMask) = 5;  % Darker pixel intensity for microhemorrhages
end


% Step 1: Create Masks for White Matter and Cerebral Cortex
whiteMatterMask = (DHP_3D == 2);   % White Matter Mask
cortexMask = (DHP_3D == 1);        % Cerebral Cortex Mask

% Step 2: Assign Pixel Values for Visualization
enhancedWhiteMatter = whiteMatterMask * 22; % Adjust White Matter Pixel Value
enhancedCortex = cortexMask * 44 ;           % Adjust Cortex Pixel Value

% Step 3: Overlay Hemorrhage Lesions (ONLY Overwriting White Matter)
DHP_micro_mild = DHP_3D; % Start with the original DHP
DHP_micro_mild(whiteMatterMask) = enhancedWhiteMatter(whiteMatterMask);  % Preserve
White Matter
DHP_micro_mild(cortexMask) = enhancedCortex(cortexMask);  % Preserve Cortex

% Overwrite lesion areas in white matter with a low intensity
lesionRegion = hemorrhageModel > 0;  % Boolean mask of hemorrhage locations
DHP_micro_mild(lesionRegion & whiteMatterMask) = hemorrhageModel(lesionRegion &
whiteMatterMask);

% Visualization of the Slices

% Set the slicing coordinates

zconstant = 77;


% Extract and display the Z-plane slice
figure;
imshow(squeeze(DHP_micro_mild(:, :, zconstant)), []);
title(sprintf('Extracted Plane at z = %d', zconstant));
```

9. Moderate Microhaemmorhages

```
% Define the DHP dimensions (Ensure this matches your DHP dataset)
xSize = 265;
ySize = 490;
zSize = 305;

% Initialize a 3D meshgrid
[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);

% Define hemorrhage lesion parameters (3 Lesions with Elliptical Shape)
lesionCenters = [
    133, 212, 60;  % Lesion 1 (Example coordinates)
    136, 213, 60;  % Lesion 2
    130, 201, 60   % Lesion 3
];

% Elliptical lesion dimensions
sigmaX = [1, 1, 2];  % Adjusted to form elliptical lesions
sigmaY = [2, 1, 3];
sigmaZ = [3, 2, 3];

% Create an empty 3D lesion model
hemorrhageModel = zeros(ySize, xSize, zSize);

% Generate microhemorrhage lesions at specified locations
for i = 1:size(lesionCenters, 1)
    cx = lesionCenters(i,1);
    cy = lesionCenters(i,2);
    cz = lesionCenters(i,3);

    % Elliptical lesion mask
    lesionMask = exp(-((x-cx).^2 / (2*sigmaX(i)^2) + (y-cy).^2 / (2*sigmaY(i)^2) +
(z-cz).^2 / (2*sigmaZ(i)^2)));

    % Apply threshold to create a hard boundary (binary mask)
    lesionMask = lesionMask > 0.5;

    % Assign a low pixel intensity (hypointense appearance)
    hemorrhageModel(lesionMask) = 5;  % Darker pixel intensity for microhemorrhages
end

% Step 1: Create Masks for White Matter and Cerebral Cortex
whiteMatterMask = (DHP_3D == 2);   % White Matter Mask
cortexMask = (DHP_3D == 1);        % Cerebral Cortex Mask

% Step 2: Assign Pixel Values for Visualization
enhancedWhiteMatter = whiteMatterMask * 22; % Adjust White Matter Pixel Value
enhancedCortex = cortexMask * 44;           % Adjust Cortex Pixel Value

% Step 3: Overlay Hemorrhage Lesions (ONLY Overwriting White Matter)
DHP_micro_moderate = DHP_3D; % Start with the original DHP
DHP_micro_moderate(whiteMatterMask) = enhancedWhiteMatter(whiteMatterMask);  %
Preserve White Matter
DHP_micro_moderate(cortexMask) = enhancedCortex(cortexMask);  % Preserve Cortex
```

```
    % Overwrite lesion areas in white matter with a low intensity
    lesionRegion = hemorrhageModel > 0;  % Boolean mask of hemorrhage locations
    DHP_micro_moderate(lesionRegion & whiteMatterMask) = hemorrhageModel(lesionRegion &
    whiteMatterMask);

    % Visualization of the Slices

    % Set the slicing coordinates

    zconstant = 60;


    % Extract and display the Z-plane slice
    figure;
    imshow(squeeze(DHP_micro_moderate(:, :, zconstant)), []);
    title(sprintf('Extracted Plane at z = %d', zconstant));
```

10. Severe Microhaemmorhages

```
% Define the DHP dimensions (Ensure this matches your DHP dataset)
xSize = 265;
ySize = 490;
zSize = 305;

% Initialize a 3D meshgrid
[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);

% Define hemorrhage lesion parameters (3 Lesions with Elliptical Shape)
lesionCenters = [
    80, 257, 87;  % Lesion 1 (Example coordinates)
    80, 262, 87;  % Lesion 1 (Example coordinates)
    115, 276, 87;  % Lesion 2
    95, 276, 87   % Lesion 3
];

% Elliptical lesion dimensions
sigmaX = [1.5, 1.5, 1.5, 1.5];  % Adjusted to form elliptical lesions
sigmaY = [1.5, 1.5, 1.5, 1.5];
sigmaZ = [1, 1, 1, 1];

% Create an empty 3D lesion model
hemorrhageModel = zeros(ySize, xSize, zSize);

% Generate microhemorrhage lesions at specified locations
for i = 1:size(lesionCenters, 1)
    cx = lesionCenters(i,1);
    cy = lesionCenters(i,2);
    cz = lesionCenters(i,3);

    % Elliptical lesion mask
    lesionMask = exp(-((x-cx).^2 / (2*sigmaX(i)^2) + (y-cy).^2 / (2*sigmaY(i)^2) + (z-
cz).^2 / (2*sigmaZ(i)^2)));

    % Apply threshold to create a hard boundary (binary mask)
    lesionMask = lesionMask > 0.5;

    % Assign a low pixel intensity (hypointense appearance)
```

```
    hemorrhageModel(lesionMask) = 5;  % Darker pixel intensity for microhemorrhages
end

% Step 1: Create Masks for White Matter and Cerebral Cortex
whiteMatterMask = (DHP_3D == 2);   % White Matter Mask
cortexMask = (DHP_3D == 1);        % Cerebral Cortex Mask

% Step 2: Assign Pixel Values for Visualization
enhancedWhiteMatter = whiteMatterMask * 22; % Adjust White Matter Pixel Value
enhancedCortex = cortexMask * 44;           % Adjust Cortex Pixel Value

% Step 3: Overlay Hemorrhage Lesions (ONLY Overwriting White Matter)
DHP_micro_severe = DHP_3D; % Start with the original DHP
DHP_micro_severe(whiteMatterMask) = enhancedWhiteMatter(whiteMatterMask);  % Preserve
White Matter
DHP_micro_severe(cortexMask) = enhancedCortex(cortexMask);  % Preserve Cortex

% Overwrite lesion areas in white matter with a low intensity
lesionRegion = hemorrhageModel > 0;  % Boolean mask of hemorrhage locations
DHP_micro_severe(lesionRegion & whiteMatterMask) = hemorrhageModel(lesionRegion &
whiteMatterMask);

% Visualization of the Slices

% Set the slicing coordinates

zconstant = 66;


% Extract and display the Z-plane slice
figure;
imshow(squeeze(DHP_micro_severe(:, :, zconstant)), []);
title(sprintf('Extracted Plane at z = %d', zconstant));
```

11. Mild Superficial Siderosis

```
% Define the DHP dimensions
xSize = 265;
ySize = 490;
zSize = 305;

% Initialize Meshgrid
[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);

% Define Microhemorrhage Lesion Parameters (Overwriting White Matter)
sigmaX_H1 = 1.5; sigmaY_H1 = 1.5; sigmaZ_H1 = 1;
centerH1 = [193, 277, 101]; % Lesion in White Matter
centerH2 = [203, 273, 101]; % Lesion in White Matter

% Define Superficial Siderosis Lesion Parameters (Overwriting Cortex)
sigmaX_S1 = 6; sigmaY_S1 = 15; sigmaZ_S1 = 1;
centerS1 = [140, 300, 101]; % Lesion in Cerebral Cortex
centerS2 = [173, 292, 101]; % Lesion in Cerebral Cortex

% Generate Microhemorrhage Lesions (STRICT Hypointense Spots)
hemorrhageModel_1 = exp(-((x-centerH1(1)).^2 / (2*sigmaX_H1^2) + (y-centerH1(2)).^2
/ (2*sigmaY_H1^2) + (z-centerH1(3)).^2 / (2*sigmaZ_H1^2))) > 0.6;
```

```
    hemorrhageModel_2 = exp(-((x-centerH2(1)).^2 / (2*sigmaX_H1^2) + (y-centerH2(2)).^2
    / (2*sigmaY_H1^2) + (z-centerH2(3)).^2 / (2*sigmaZ_H1^2))) > 0.6;

    % Generate Superficial Siderosis Lesions (STRICT Hypointense Spots)
    siderosisModel_1 = exp(-((x-centerS1(1)).^2 / (2*sigmaX_S1^2) + (y-centerS1(2)).^2 /
    (2*sigmaY_S1^2) + (z-centerS1(3)).^2 / (2*sigmaZ_S1^2))) > 0.6;
    siderosisModel_2 = exp(-((x-centerS2(1)).^2 / (2*sigmaX_S1^2) + (y-centerS2(2)).^2 /
    (2*sigmaY_S1^2) + (z-centerS2(3)).^2 / (2*sigmaZ_S1^2))) > 0.6;

    % Set Lesion Intensity (Darker Appearance)
    lesionIntensity = 5;
    hemorrhageModel_1 = hemorrhageModel_1 * lesionIntensity;
    hemorrhageModel_2 = hemorrhageModel_2 * lesionIntensity;
    siderosisModel_1 = siderosisModel_1 * lesionIntensity;
    siderosisModel_2 = siderosisModel_2 * lesionIntensity;

    % Step 1: Create Masks for White Matter and Cerebral Cortex
    whiteMatterMask = (DHP_3D == 2);   % White Matter Mask
    cortexMask = (DHP_3D == 1);        % Cerebral Cortex Mask

    % Step 2: Preserve Original White Matter and Cortex Intensities
    DHP_sid_mild = DHP_3D;  % Start with original DHP

    % Apply White Matter Mask (Fixed Intensity)
    DHP_sid_mild(whiteMatterMask) = 34;

    % Apply Cortex Mask (Fixed Intensity)
    DHP_sid_mild(cortexMask) = 65;

    % Step 3: Overlay Microhemorrhages (ONLY Overwriting White Matter)
    DHP_sid_mild(whiteMatterMask & hemorrhageModel_1) = lesionIntensity;
    DHP_sid_mild(whiteMatterMask & hemorrhageModel_2) = lesionIntensity;

    % Step 4: Overlay Superficial Siderosis (ONLY Overwriting Cerebral Cortex)
    DHP_sid_mild(cortexMask & siderosisModel_1) = lesionIntensity;
    DHP_sid_mild(cortexMask & siderosisModel_2) = lesionIntensity;

    % Set the slicing coordinates

    zconstant = 101;


    % Extract and display the Z-plane slice
    figure;
    imshow(squeeze(DHP_sid_mild(:, :, zconstant)), []);
    title(sprintf('Extracted Plane at z = %d', zconstant));
```

12. Moderate Superficial Siderosis

```
% Define the DHP dimensions
xSize = 265;
ySize = 490;
zSize = 305;

% Initialize Meshgrid
[x, y, z] = meshgrid(1:xSize, 1:ySize, 1:zSize);
```

```
% Define Microhemorrhage Lesion Parameters (Overwriting White Matter)
sigmaX_H1 = 1.2; sigmaY_H1 = 1.2; sigmaZ_H1 = 1;
centerH1 = [87, 220, 53]; % Lesion in White Matter


% Define Superficial Siderosis Lesion Parameters (Overwriting Cortex)
sigmaX_S1 = 20; sigmaY_S1 = 10; sigmaZ_S1 = 1;
centerS1 = [90, 265, 53]; % Lesion in Cerebral Cortex
centerS2 = [100, 273, 53]; % Lesion in Cerebral Cortex

% Generate Microhemorrhage Lesions (STRICT Hypointense Spots)
hemorrhageModel_1 = exp(-((x-centerH1(1)).^2 / (2*sigmaX_H1^2) + (y-centerH1(2)).^2 /
(2*sigmaY_H1^2) + (z-centerH1(3)).^2 / (2*sigmaZ_H1^2))) > 0.6;


% Generate Superficial Siderosis Lesions (STRICT Hypointense Spots)
siderosisModel_1 = exp(-((x-centerS1(1)).^2 / (2*sigmaX_S1^2) + (y-centerS1(2)).^2 /
(2*sigmaY_S1^2) + (z-centerS1(3)).^2 / (2*sigmaZ_S1^2))) > 0.6;
siderosisModel_2 = exp(-((x-centerS2(1)).^2 / (2*sigmaX_S1^2) + (y-centerS2(2)).^2 /
(2*sigmaY_S1^2) + (z-centerS2(3)).^2 / (2*sigmaZ_S1^2))) > 0.6;

% Set Lesion Intensity (Darker Appearance)
lesionIntensity = 5;
hemorrhageModel_1 = hemorrhageModel_1 * lesionIntensity;
siderosisModel_1 = siderosisModel_1 * lesionIntensity;
siderosisModel_2 = siderosisModel_2 * lesionIntensity;

% Step 1: Create Masks for White Matter and Cerebral Cortex
whiteMatterMask = (DHP_3D == 2);   % White Matter Mask
cortexMask = (DHP_3D == 1);        % Cerebral Cortex Mask

% Step 2: Preserve Original White Matter and Cortex Intensities
DHP_sid_moderate = DHP_3D;  % Start with original DHP

% Apply White Matter Mask (Fixed Intensity)
DHP_sid_moderate(whiteMatterMask) = 34;

% Apply Cortex Mask (Fixed Intensity)
DHP_sid_moderate(cortexMask) = 65;

% Step 3: Overlay Microhemorrhages (ONLY Overwriting White Matter)
DHP_sid_moderate(whiteMatterMask & hemorrhageModel_1) = lesionIntensity;


% Step 4: Overlay Superficial Siderosis (ONLY Overwriting Cerebral Cortex)
DHP_sid_moderate(cortexMask & siderosisModel_1) = lesionIntensity;
DHP_sid_moderate(cortexMask & siderosisModel_2) = lesionIntensity;

% Set the slicing coordinates

zconstant = 77;


% Extract and display the Z-plane slice
figure;
imshow(squeeze(DHP_sid_moderate(:, :, zconstant)), []);
title(sprintf('Extracted Plane at z = %d', zconstant));
```

13. Load PGM File Function

```
function writePGM(filename, img)
    img = uint8(img);
    [rows, cols] = size(img);
    maxVal = max(img(:)); % detect actual maximum
    fid = fopen(filename, 'w');
    fprintf(fid, 'P2\n%d %d\n%d\n', cols, rows, maxVal); % write actual max
    for i = 1:rows
        fprintf(fid, '%d ', img(i,:));
        fprintf(fid, '\n');
    end
    fclose(fid);
end
```